Tariff Bindings and the Dynamic Formation of Preferential Trade Agreements

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Abstract

We investigate the impact of multilateral tariff binding liberalization on the equilibrium extent of Preferential Trade Agreement (PTA) formation. When forming Free Trade Agreements (FTAs) under relatively high tariff bindings, a myopic free riding incentive of FTA non-members constrains FTA formation. Thus, tariff binding liberalization can facilitate FTA expansion to global free trade. However, when forward looking countries do not value this myopic free riding incentive, tariff binding liberalization can impede FTA expansion to global free trade by effectively increasing tariff concessions given by the FTA non-member so that FTA members exclude the FTA non-member from subsequent FTA expansion. When countries choose between forming FTAs and Custom Unions (CUs), the equilibrium emergence of CUs can prevent (cannot facilitate) global free trade that would otherwise occur (not occur) through FTAs. In contrast, the equilibrium emergence of FTAs can facilitate (cannot prevent) global free trade that would otherwise not occur (occur) through CUs.

Keywords: Tariff bindings, Preferential Trade Agreement, Free Trade Agreement, Customs Union, global free trade, dynamic.

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1 Introduction

Since 1948, the General Agreement on Tariffs and Trade (GATT), codified by the creation of the World Trade Organization (WTO) in 1995, has governed global trade liberalization. The pillar of GATT is the Most Favored Nation (MFN) principle. This principle of nondiscrimination requires a country levy the same tariff, the so-called MFN tariff, on other GATT/WTO members. Most prominently, various GATT negotiation rounds from the 1947 Geneva Round through to the 1994 Uruguay Round generated substantial MFN tariff concessions with countries committing to "tariff bindings" which cap their maximum applied MFN tariff. While less prominent, a large subset of WTO members have also committed to zero tariff bindings on a range of Information Technology products via the Information Technology Agreement (ITA). The set of ITA countries and products has grown over time and this plurilateral type of agreement for product-specific tariff binding concessions is the prototype for current negotiations on an Environmental Goods Agreement. Ultimately, MFN tariff concessions have been and continue to be a cornerstone of global trade liberalization.

Nevertheless, in direct contrast to the central pillar of GATT being the non-discriminatory MFN principle, GATT also allows discriminatory liberalization through Preferential Trade Agreements (PTAs). Specifically, GATT Article XXIV allows countries to eliminate tariffs between themselves if they do not raise barriers on other countries. Countries can either form Free Trade Agreements (FTAs) and maintain sovereign discretion over their "external tariffs" on non-members or Customs Unions (CUs) and set a common external tariff on non-members. Although relatively rare in the pre-Uruguay Round world, PTAs have proliferated in the post-Uruguay Round world. The tension between these discriminatory and non-discriminatory modes of liberalization spawned and, through failure of the current Doha Round, has sustained a large literature. In particular, this literature focuses on understanding how the discriminatory nature of PTAs impacts the degree of global trade liberalization that would otherwise arise in terms of non-discriminatory MFN tariffs set by countries either individually or through global negotiation rounds.

While the literature has long understood that PTAs can impact the extent of MFN tariff liberalization and hence the ultimate degree of global liberalization, the literature has largely ignored that the extent of MFN tariff liberalization can impact PTA formation and hence the ultimate degree of global tariff liberalization. This is despite the fact that real world MFN tariff liberalization often takes the form of tariff binding reductions which directly impact the incentives for PTA formation through changing the MFN tariffs that countries face in the absence of a PTA. This observation raises the first key question of our paper: how does a continual reduction in tariff bindings across the world impact the role of FTAs and CUs

in helping or hindering the attainment of global free trade?

The classic building bloc-stumbling bloc issue in the PTA literature is typically viewed as a comparison between PTA formation and multilateral liberalization in terms of whether PTA formation improves or hurts the prospects of global free trade relative to the outcome under multilateral liberalization. However, in practice, multilateral liberalization happens in the short bursts in the background whether it be through various global rounds of negotiations or plurilateral agreements such as the Information Technology Agreement. In contrast, countries are continuously forming PTAs on the surface of the global trade policy landscape and, in doing so, choosing whether to form FTAs or CUs. Thus, an alternative view of the building bloc-stumbling bloc issue, and perhaps a more informative view, would ask the following question: does a given type of PTA improve or hurt the prospects of global free trade relative to the outcome if this type of PTA was banned? This is the second key question of our paper.

We build a three country dynamic model of PTA formation where countries form PTAs over time. For our underlying trade model, we use a competing exporters model where each country imports one good from the other two countries and production technologies exhibit increasing costs. To focus on the impact of continual reductions in tariff bindings, our model features symmetric countries and a symmetric tariff binding. In turn, we investigate how equilibrium PTA formation, and specifically the attainment of global free trade, changes with continual reductions in the symmetric tariff binding.

To address how continual reductions in the tariff binding impact the role of FTAs in helping or hindering the attainment of global free trade, we analyze an "FTA game" where countries can only form FTAs. With relatively high tariff binding, the key intuition revolves around free riding by the FTA outsider. Intuitively, despite facing discrimination, the FTA outsider benefits from the well known tariff complementarity phenomena whereby FTA members lower their MFN tariff upon FTA formation. Indeed, the FTA outsider actually becomes worse off upon a subsequent FTA with an FTA insider. In this sense, the FTA outsider holds a *myopic free riding incentive*. However, by eliminating the discrimination faced in both FTA insider markets, the FTA outsider benefits from global free trade. The tension between this farsighted benefit of global free trade and the myopic free riding incentive generates a dynamic trade-off. A sufficiently patient FTA outsider willingly becomes the spoke because of the farsighted benefit of global free trade. But, a sufficiently myopic FTA outsider holds a *dynamic free riding incentive* in that it refuses any subsequent FTA formation because the myopic free riding incentive dominates the farsighted benefit of global free trade.

However, continual reductions in the global tariff binding eventually eliminate the myopic

free riding incentive by constraining the FTA outsider's ability to levy its optimal tariffs on the FTA insiders. With a relatively low tariff binding, the key intuition revolves around the FTA insiders holding an *exclusion incentive*. Intuitively, the FTA insiders have already extracted tariff concessions from the FTA outsider through the relatively low tariff binding and benefit from excluding the FTA outsider from global free trade to protect their preferential access as FTA insiders. Nevertheless, an FTA insider benefits myopically from forming its own FTA with the FTA outsider and thus becoming the "hub" with preferential access to both of the other "spoke" countries. The tension between this myopic benefit of becoming the hub and the farsighted nature of the exclusion incentive generates a dynamic trade off. A sufficiently myopic FTA insider becomes the hub because of the benefits associated with preferential access to both spoke markets. But, a sufficiently patient FTA insider holds a *dynamic exclusion incentive* in that it refuses any subsequent FTA formation because of the farsighted benefit of not precipitating expansion to global free trade.

Two main insights emerge when analyzing how continual tariff binding reductions change how FTA formation helps or hinders the attainment of global free trade. First, when high tariff bindings prevent FTA expansion to global free trade, continual reductions in the tariff binding facilitate FTA formation that attains global free trade. Specifically, suppose the dynamic free riding incentive prevents global free trade under a sufficiently high tariff binding noting that this requires sufficient myopia. Then, continual reductions in the tariff binding eventually eliminate the dynamic free riding incentive and, despite generating an exclusion incentive, FTA insiders do not hold a *dynamic* exclusion incentive because the dynamic exclusion incentive requires sufficient patience. In turn, FTA formation yields global free trade.

The second insight is that when high tariffs bindings facilitate FTA expansion to global free trade, continual reductions in tariff bindings change the nature of FTA formation so that FTA expansion no longer facilitates but can rather impede FTA expansion to global free trade. This insight may be especially relevant given the literature documents FTA formation typically takes many years. As such, tariff binding reductions may occur before FTA expansion has time to reach global free trade. In this case, the FTA outsider does not hold a *dynamic* free riding incentive under the initially high tariff binding because of sufficient patience. However, continual tariff binding reductions create an exclusion incentive and FTA insiders may hold a *dynamic* exclusion incentive since this also requires sufficient patience. In this case, FTA formation would eventually yield global free trade under the initial and relatively high tariff binding, but continual reductions in the tariff binding generate a dynamic exclusion incentive for FTA insiders who refuse any subsequent FTA formation with the FTA outsider.

Unlike our "FTA game" which features tensions between myopic and farsighted incentives, our "CU game" is much simpler. While the CU non-member always benefits from CU expansion to global free trade, CU insiders want to exclude the CU non-member from such expansion when the tariff binding is relatively low. As for FTA insiders, a relatively low tariff binding implies the CU members have extracted substantial tariff concessions from the CU non-member without giving concessions to the CU non-member via CU expansion. Thus, CU formation proceeds to global free trade only for relatively high tariff bindings.

As described above, the second key question of our paper is whether a given type of PTA (i.e. FTAs or CUs) improves or hurts the prospects of global free trade relative to the outcome if this type of PTA was banned? To this end, we analyze our "PTA game" where countries can choose between FTA and CU formation and we introduce the terminology of a "PTA stumbling bloc" and a "PTA building bloc". We say that CUs (FTAs) are a "PTA stumbling bloc" if CU (FTA) formation emerges in equilibrium and does not lead to global free trade yet FTA (CU) formation would lead to global free trade in the absence of CUs (FTAs). Similarly, we say that CUs (FTAs) are a "PTA building bloc" if CU (FTA) formation emerges in equilibrium and leads to global free trade yet FTA (CU) formation would not lead to global free trade in the absence of CUs (FTAs).

Whether FTAs or CUs emerge in equilibrium revolves around a dynamic trade off. On one hand, CU formation confers coordination benefits on CU members. From a myopic perspective, CU members benefit from coordination of their external tariffs internalizing the negative intra-PTA externality of tariff complementarity. From a farsighted perspective, CU members benefit from the implication that coordination of their external tariffs confers veto power on each CU member regarding subsequent CU expansion to global free trade. This veto power is valuable in the presence of a CU exclusion incentive because, in contrast, each FTA member can form their own subsequent FTA with the FTA non-member and thus precipitate FTA expansion to global free trade. On the other hand, the ability of each FTA member to form their own subsequent FTA with the non-member confers an FTA flexibility benefit on FTA members. This FTA flexibility benefit is valuable because it allows a member of an initial FTA to then become the "hub" and have sole preferential access with the other two "spoke" countries. In general, this FTA flexibility dominates the myopic and farsighted CU coordination benefits, and hence FTA formation emerges in equilibrium, when the discount factor lies in an intermediate range.

Ultimately, FTAs and CUs play very different roles in helping or hurting the prospects for global free trade. On one hand, FTAs can be PTA building blocs but never PTA stumbling blocs. That is, when FTA formation emerges in equilibrium then (i) it can lead to global free trade when CU formation would not but (ii) if it does not lead to global free trade then

neither does CU formation. On the other hand, CUs can be PTA stumbling blocs but never PTA building blocs. That is, when CU formation emerges in equilibrium then (i) there are conditions where CU expansion will not, but FTA expansion will, lead to global free trade and (ii) when CU formation leads to global free trade then so does FTA formation. In particular, this PTA building bloc role of FTAs and PTA stumbling bloc role of CUs emerges for an intermediate degree of tariff bindings. This suggests that the asymmetric impact of FTAs versus CUs may have become more prevalent with the phase in of Uruguay Round multilateral tariff binding liberalization and plurilateral tariff binding agreements like the Information Technology Agreement.

Surprisingly few papers have addressed the impact of multilateral tariff liberalization on the extent of PTA formation (Freund and Ornelas (2010)). Ethier (1998) argues PTA formation is a benign consequence of multilateralism whereby countries left out of earlier rounds of multilateral tariff liberalization utilize PTA formation. In a repeated game setting, Freund (2000) shows how multilateral tariff liberalization can make an FTA "self-enforcing". Like our paper, Nken and Yildiz (2017) find tariff binding liberalization weakens the free riding incentive of FTA non-members. Because of this, like Freund (2000), multilateral tariff liberalization facilitates FTA expansion. In contrast to these positive views regarding multilateral tariff liberalization, Lake and Roy (2017) show how endogenous determination of multilateral tariff bindings generates an FTA exclusion that otherwise would not exist. Like our paper, relatively tight multilateral tariff bindings deliver tariff concessions from FTA non-members to FTA members without FTA members needing to form FTAs with non-members. Ultimately, multilateralism acts as a stumbling bloc to global free trade for Lake and Roy (2017).

Exclusion incentives for PTA members and free riding incentives for PTA non-members have long been important for theoretical analyses of PTA formation and their role in the attainment of global free trade.[1] However, a given analysis typically only relies on an exclusion incentive of PTA members or only relies on a free riding incentive of PTA non-members. Relying on free riding incentives, Saggi and Yildiz (2010) show the incentive to free ride on FTA formation is weaker than the incentive to free ride on MFN liberalization of other countries and, hence, FTAs can be a "building bloc" to global free trade. In contrast, Saggi et al. (2013) rely on an exclusion incentive and show the incentive of CU members to exclude the non-member can be stronger than the MFN free riding incentive and, hence, CUs can be a "stumbling bloc" to global free trade. Based on these two papers, Maggi (2014) hypothesizes

[1] Indeed, the seminal contributions of Levy (1997) and Krishna (1998) relied on exclusion incentives. Ornelas (2005) represents an early analysis relying on a free riding incentive. In a recent survey chapter, Lake and Krishna (2018) emphasizes the role played by exclusion incentives and free riding incentives in the theoretical PTA literature.

that CU formation constrains the prospects of global free trade but argues such a conclusion requires a model where countries endogenously choose between CU and FTA formation.

Our paper addresses Maggi's hypothesis by developing a model where countries endogenously choose between FTA and CU formation. In doing so, we build a model where PTA members can hold exclusion incentives and PTA non-members can hold free riding incentives. Rather than tie the existence of these incentives to country asymmetries, which are somewhat nebulous in terms of real world application of theoretical PTA models, we tie them to tariff binding liberalization which has concrete real world interpretations as we discussed above. Further, we build a dynamic model rather than a static model because static models often have difficulty in generating the equilibrium emergence of FTAs; for example, Missios et al. (2016) show that countries always choose CU formation over FTA formation in their static model. As in Lake and Yildiz (2016) and Lake (2018), the dynamic setting allows the dynamic trade off between the FTA flexibility benefit and the CU coordination benefits to determine whether FTAs or CUs emerge in equilibrium. Ultimately, we confirm Maggi's hypothesis based on the different models of Saggi and Yildiz (2010) and Saggi et al. (2013): in our own terminology, FTAs act as a "PTA building bloc" but CUs act as a "PTA stumbling bloc".

The paper proceeds as follows. Section 2 describes our model, both the underlying trade model and the game theoretic model of PTA formation. Section 3 analyzes the equilibrium outcome of the "FTA game" where countries can only form FTAs. Sections 3-5 analyze the equilibrium outcome of PTA formation with Section 3 analyzing the "FTA game" where countries can only form FTAs, Section 4 analyzing the "CU game" where countries can only form CUs and Section 5 analyzing the "PTA game" where countries endogenously choose between FTAs and CUs. Finally, Section 6 concludes. The Appendix contains all proofs.

2 Model

2.1 Underlying trade model

We use a three-country competing exporters version of the Horn et al. (2010) two-country model. Our model consists of three countries a, b and c; three (non-numeraire) goods A, B and C; and, a numeraire good v_0. Where appropriate, we hereafter use $z = i, j, k$ as generic notation for countries and $Z = I,\ J,\ K$ as generic notation for non-numeraire goods.

On the demand side, the representative citizen's utility function is quasi-linear

$$U(\mathbf{v}, v_0) = u(\mathbf{v}) + v_0$$

with $u(\mathbf{v})$ both quadratic and additively separable in the vector of non-numeraire good consumption $\mathbf{v}$. Thus, a representative consumer from country i has demand for good Z of

$$d_i^Z(p_i^Z) = \alpha - p_i^Z \tag{1}$$

where p_i^Z denotes the price of good Z in country i.[2] In turn, country i's consumer surplus from good Z is

$$CS_i^Z(p_i^Z) = u_i^Z[d_i^Z(p_i^Z)] - p_i^Z d_i^Z(p_i^Z) = \frac{1}{2}\left(\alpha - p_i^Z\right)^2. \tag{2}$$

On the supply side, labour (l) is the only factor of production. The numeraire good is produced one-for-one from labor.[3] In contrast, production Q_i^Z of a non-numeraire good Z in country i follows the diminishing returns production function:

$$Q_i^Z = \sqrt{2\lambda_i^Z l_i^Z}$$

where l_i^Z denotes the labor employed in the production of good Z in country i and λ_i^Z depends on the structure on comparative advantage. In turn, the supply function for good Z in country i is

$$s_i^Z(p_i^Z) = \lambda_i^Z p_i^Z \tag{3}$$

and the producer surplus for good Z in country i is:

$$PS_i^Z(p_i^Z) = \int s_i^Z(p_i^Z)dp_i^Z = \frac{1}{2}\lambda_i^Z(p_i^Z)^2 \tag{4}$$

We assume a symmetric structure of comparative advantage across countries. Specifically, $\lambda_i^I = 1$ while $\lambda_i^Z = 1+\lambda$ for $Z \neq I$ so that each country i has a comparative advantage in the two goods $Z \neq I$ while having a comparative disadvantage in good I. Thus, countries j and k compete with each other when exporting good I to country i. That is, a single country imports each non-numeraire good and two *competing exporters* serve this destination market.

No arbitrage conditions link the prices of non-numeraire goods across countries and world market clearing conditions for non-numeraire goods deliver equilibrium prices. Ruling out prohibitive tariffs and letting t_{iz} denote country i's tariff on imports of good I from country $z = j, k$, the no-arbitrage condition for good I implies:

$$p_i^I = p_j^I + t_{ij} \ = p_k^I + t_{ik} \tag{5}$$

[2] Our focus on tariffs as the only policy instrument implies there is no distinction between the prices faced by consumers and producers in a country.

[3] This pins wages to 1 given we also assume the supply of labor is large enough to ensure positive production of the numéraire good.

World market clearing for good I requires

$$m_i^I = \sum_{z \neq i} x_z^I \tag{6}$$

where $m_i^I = d(p_i^I) - s_i^I(p_i^I)$ denotes country i's imports of good I and $x_z^I = s_z^I(p_z^I) - d(p_z^I)$ denotes the exports of good I by country $z = j, k$. Specifically, given (1) and (3), we have

$$\begin{aligned} m_i^I &= \alpha - 2p_i^I & (7) \\ x_z^I &= (1+\lambda)p_z^I - \left[\alpha - p_z^I\right]. & (8) \end{aligned}$$

Substituting the trade flows (7)-(8) and the no-arbitrage condition (5) into the world market clearing condition (6) yields the equilibrium prices of good I in the importing country i and an exporting country j:

$$p_i^I = \frac{3\alpha + (2+\lambda)\sum_{z \neq i} t_{iz}}{2(3+\lambda)} \text{ and } p_j^I = \frac{3\alpha + (2+\lambda)t_{ik} - (4+\lambda)t_{ij}}{2(3+\lambda)}. \tag{9}$$

These equilibrium prices show how tariffs and the degree of comparative advantage impact prices and trade flows. Given equilibrium prices, country j's exports of good I to country i are

$$x_j^I = \frac{\alpha\lambda\left[t_{ik}(\lambda^2 + 4(1+\lambda)) - t_{ij}(\lambda^2 + 2(3\lambda+4))\right]}{2\lambda+3}. \tag{10}$$

Three observations follow from (10). First, a stronger degree of comparative advantage (i.e. higher λ) increases trade flows between countries (i.e. higher x_z^I for $z = j, k$) and lowers prices across the world (i.e. lower p_i^I and p_z^I for $z = j, k$). Second, while a fraction $\frac{2+\lambda}{2\lambda+6}$ of country i's tariff on country j passes through to a higher local price p_i^I, the remaining fraction $\frac{4+\lambda}{2\lambda+6}$ passes through to a lower local price p_j^I which represents a terms of trade improvement for country i vis-à-vis country j. Third, the tariff on country j also diverts trade between the competing exporters: country k's exports to country i rise on account of the higher price p_k^I while country j's exports to country i fall on account of the lower price p_j^I.

Finally, given the partial equilibrium nature of the model, trade policy has welfare implications only for the protected non-numeraire goods. A country's welfare is defined as the sum of consumer surplus, producer surplus, and tariff revenue over all such goods:

$$w_i = \sum_Z \left(CS_i^Z + PS_i^Z\right) + \sum_{z \neq i} t_{iz} x_z^I \tag{11}$$

Using (9) with (2), (4) and (10) yields closed form expressions for national welfare as a

function of tariffs and the degree of comparative advantage.[4]

2.1.1 Optimal applied tariffs

In this section, we focus on optimal applied tariffs and ignore tariff bindings. Section 2.1.2 considers the implications of tariff bindings for the applied tariffs imposed by countries.

Before discussing optimal applied tariffs, we introduce terminology that describes the network of PTAs. A growing number of papers in the recent PTA literature (e.g. Goyal and Joshi (2006), Furusawa and Konishi (2007), Zhang et al. (2013), Lake and Yildiz (2016) Lake (2017)) borrows terminology from the network literature by viewing countries as nodes on a "graph" and edges between nodes as bilateral "links" between countries. The graph g is then described by the set of bilateral links between players and we henceforth refer to the graph as the network.

Figure 1 describes the possible networks among three countries. The "empty network" $g_{\varnothing}$ emerges in the absence of any PTAs. g_{ij}^{FTA} describes the "FTA insider-outsider network" where the FTA non-member k is the "FTA outsider" and the FTA members i and j are "FTA insiders". Analogously, g_{ij}^{CU} describes the "CU insider-outsider network" where the CU non-member k is the "CU outsider" and the CU members i and j are "CU insiders". g_i^H denotes the "hub-spoke" network where countries j and k are "spokes" and each have an FTA with country i who is the "hub". g^{FT} describes the "free trade network" where all countries are linked through FTAs or CUs.

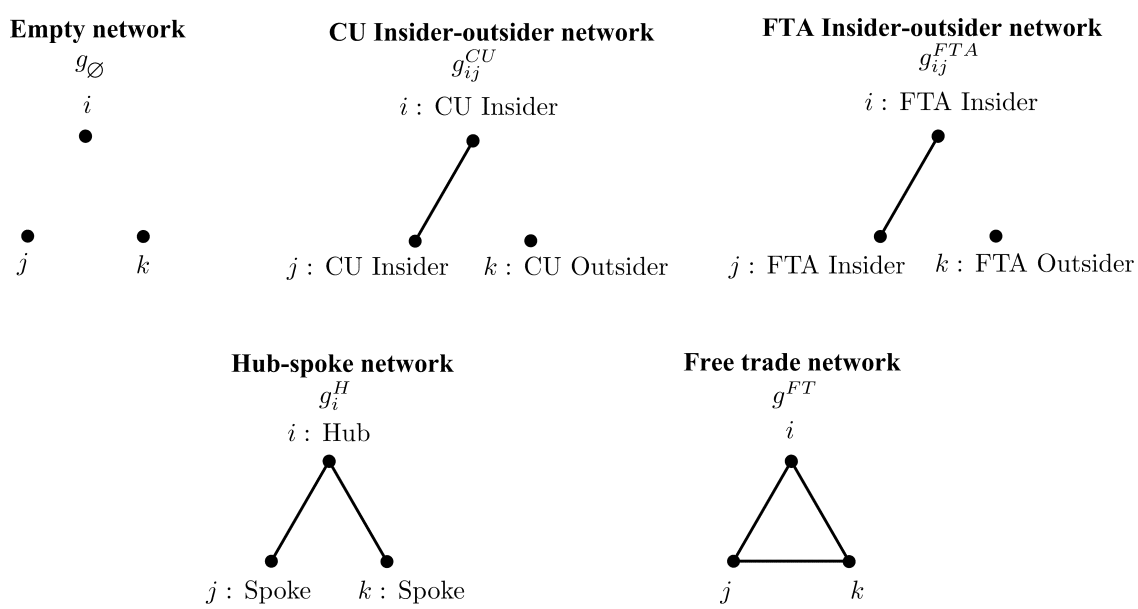

Figure 1: Network positions

[4]The Appendix contains closed form welfare expressions.

In the absence of having any PTAs, country i imposes a non-discriminatory tariff on its trading partners (in accordance with GATT Article I). Thus, letting $t_{ij}(g_\varnothing) = t_{ik}(g_\varnothing) = t_i(g_\varnothing)$, country i's optimal MFN tariff is:

$$t_\varnothing \equiv \underset{t_i(g_\varnothing)}{\arg\max} w_i(g_\varnothing) = \frac{\alpha\lambda}{2(2+\lambda)(4+\lambda)}. \tag{12}$$

By increasing its volume of imports, a country's optimal tariff increases with its own market size (α) and the exporters' degree of comparative advantage (λ).

Upon FTA formation, member countries remove their internal tariffs on each other and impose an individually optimal external tariff on the non-member. Under a single FTA, say between i and j, the optimal external tariff of each FTA insider is

$$t_{FTA} \equiv \underset{t_{ik}\left(g_{ij}^{FTA}\right)}{\arg\max} w_i(g_{ij}^{FTA}) = \frac{\alpha\lambda}{2(2+\lambda)(2\lambda^2+13\lambda+22)}. \tag{13}$$

Note that the separability across markets implies the optimal MFN tariff of an FTA non-member remains unchanged. Thus, an FTA outsider's optimal MFN tariff remains unchanged relative to the empty network, $t_{ki}\left(g_{ij}^{FTA}\right) = t_{kj}\left(g_{ij}^{FTA}\right) = t_k(g_\varnothing)$, and the optimal MFN tariff of spokes equal those of FTA insiders, $t_{jk}\left(g_i^H\right) = t_{jk}\left(g_{ij}^{FTA}\right)$. Comparing t_{FTA} and $t_\varnothing$ reveals that $t_\varnothing > t_{FTA}$ and, hence, FTA insiders practice "tariff complementarity" in that FTA formation induces them to lower their tariff on the FTA outsider. This tariff complementarity becomes deeper as the degree of exporter comparative advantage rises: $\frac{\partial(t_\varnothing - t_{FTA})}{\partial\lambda} > 0$.[5]

CU members also practice tariff complementarity. Like FTA insiders, CU insiders remove tariffs on each other: $t_{ij}\left(g_{ij}^{CU}\right) = t_{ji}\left(g_{ij}^{CU}\right) = 0$. However, unlike FTA insiders, CU insiders coordinate their external tariffs. By maximizing their joint welfare, the optimal external tariff for CU insiders is

$$t_{CU} \equiv \underset{t_{ik}\left(g_{ij}^{CU}\right)}{\arg\max}\ w_i(g_{ij}^{CU}) + w_j(g_{ij}^{CU}) \text{ subject to } t_{jk}\left(g_{ij}^{CU}\right) = t_{ik}\left(g_{ij}^{CU}\right) \tag{14}$$

$$= \frac{\alpha\lambda}{(\lambda+2)(3\lambda+10)}. \tag{15}$$

While CU insiders also practice tariff complementarity, i.e. $t_\varnothing > t_{CU}$, the degree of the tariff complementarity is less than the one of FTA insiders: $t_{CU} > t_{FTA}$. Intuitively, when

[5] Detailed discussions of tariff complementarity can be found in Bagwell and Staiger (1997b, 1999) and Saggi and Yildiz (2009) while Estevadeordal et al. (2008) provide empirical support. Among others, Bond et al. (2004) show the emergence of tariff complementarity in a simple general equilibrium models of trade agreements.

setting external tariffs individually, each FTA member does not consider the negative externality imposed on the export surplus of its FTA partner by lowering its tariff on the FTA non-member. By coordinating their external tariffs, CU members internalize this negative externality and thereby benefit from tariff coordination.[6]

2.1.2 Implications of tariff bindings

Naturally, countries cannot raise their applied tariff above the tariff binding they have agreed during multilateral negotiations. In this paper, we take the tariff bindings from multilateral negotiations as exogenous. Further, given our symmetric countries, we consider a symmetric tariff binding τ.

Given the optimal tariffs in Section 2.1.1 and an exogenous tariff binding τ, Figure 2 illustrates four possible ranges. The last region in Figure 2 is the "No binding region". Here, regardless of PTA formation, the tariff binding τ exceeds the optimal applied tariff of all countries: $\tau > t_{\varnothing} > t_{CU} > t_{FTA}$. Thus, the tariff binding never binds. This is the range typically considered by the existing PTA literature that ignores the implications of multilaterally negotiated tariff bindings.

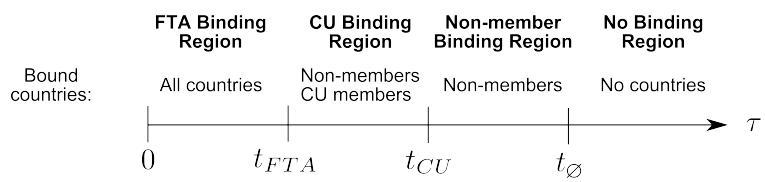

Figure 2: Tariff binding regions

However, the tariff binding binds once $\tau < t_{\varnothing}$. In the "Non-member binding region", $t_{CU} < \tau < t_{\varnothing}$ which says the tariff binding binds the applied tariffs of the FTA outsider and CU outsider who thus levy an applied tariff of τ that lies below their optimal applied tariff $t_{\varnothing}$. However, this tariff binding τ does not bind any other countries.

The tariff binding binds the applied tariffs of CU insiders (in addition to FTA and CU outsiders) once $t_{FTA} < \tau < t_{CU}$. That is, CU insiders levy an applied tariff τ that lies below their optimal applied tariff t_{CU}. Finally, once in the "FTA binding region" of $\tau < t_{FTA}$, the tariff binding now binds the applied tariffs of all countries who all levy an applied tariff τ below their optimal applied tariff.

[6] In competing exporter models, the coordination of external tariffs by CU members internalizes the negative intra-PTA externality of tariff complementarity. However, in competing importers model like Bagwell and Staiger (1997a), the coordination of external tariffs by CU members benefits CU members because they pool their market power and extract larger terms of trade gains from non-members.

Network at beginning of current period	Network at end of current period
$g_{\varnothing}$	$g_{\varnothing}, g_{ij}^{FTA}, g_{ik}^{FTA}, g_{jk}^{FTA}, g_{ij}^{CU}, g_{ik}^{CU}, g_{jk}^{CU}$
g_{ij}^{FTA}	$g_{ij}^{FTA}, g_{i}^{H}, g_{j}^{H}$
g_{ij}^{CU}	g_{ij}^{CU}, g^{FT}
g_{i}^{H}	g_{i}^{H}, g^{FT}
g^{FT}	g^{FT}

Table 1: Feasible network transitions

2.2 Strategies and equilibrium concept

Our dynamic model follows Lake and Yildiz (2016) and Lake (2018), and is similar to Seidmann (2009). We assume at most one PTA can form in a period and that PTAs formed in previous periods are binding and hence cannot be severed.[7,8] Given a network g_{t-1} at the beginning of the current period t, we refer to the current period t as the subgame at g_{t-1}. Table 1 illustrates the current period feasible transitions $g_{t-1} \rightarrow g_t$ where g_t denotes the network at the end of the current period.

Note that a network remains permanently when one of two conditions occur, and this happens from no later than the third period. First, when no agreement forms in a given period, the assumption below of Markov strategies implies no agreement will form in any subsequent period. Second, once global free trade emerges, the assumption that previously formed agreements cannot be severed implies global free trade remains forever. Because the network remains unchanged from no later than the third period onwards, we let the last network in a path of networks denote the network that remains forever: for example, $g_{t-1} \rightarrow g_t \rightarrow g_{t+1}$ describes the path of networks that begins at g_{t-1} and then passes through g_t and then remains at g_{t+1} forever; alternatively, the *path* of networks $g_{\varnothing} \rightarrow g_{\varnothing}$ indicates that no PTA ever forms.

Ultimately, countries have preferences over paths of PTA networks with β denoting the discount factor. To this end, given a network g, we assume country i's one period payoff is its national welfare $w_i(g)$. Continuation payoffs capture country preferences over paths of PTA networks. Given a feasible transition $g_{t-1} \rightarrow g_t$ from the network g_{t-1} at the beginning of the current period to the network g_t at the end of the current period, the context will often make clear the path of networks $g_{t+1} \rightarrow g_{t+2} \rightarrow ...$ that follow this current period transition. Thus,

[7] Ornelas (2008) and Ornelas and Liu (2012), among others, argue the binding nature of trade agreements is pervasive in the literature and, nothwithstanding Brexit, realistic in terms of real world observation. They also argue the assumption of binding agreements can represent a reduced form for more structural justifications such as sunk costs (see McLaren (2002) and, for empirical support, Freund and McLaren (1999)).

[8] Because, negotiations often take many years to complete (e.g. despite not being signed until 1992, Odell (2006, p.193) documents NAFTA negotiations dating back to 1986; also see Mölders (2012, 2015) and Freund and McDaniel (2016)), we essentially interpret a period as the required time to negotiate an agreement.

we simply let $V_i(g_t)$ represent the continuation payoff from the path of networks $g_{t-1} \rightarrow g_t \rightarrow g_{t+1} \rightarrow g_{t+2} \rightarrow ....$ For concreteness, suppose the network at the beginning of the current period is the empty network $g_\varnothing$. Then, country i's continuation payoff from the path of networks $g_\varnothing \rightarrow g_{ij}^{CU} \rightarrow g^{FT}$ is $V_i\left(g_{ij}^{CU}\right) = w_i\left(g_{ij}^{CU}\right) + \frac{\beta}{1-\beta} w_i\left(g^{FT}\right)$. Alternatively, given the empty network $g_\varnothing$ remains forever if no PTA forms in the current period, country i's continuation payoff from the path of networks $g_\varnothing \rightarrow g_\varnothing$ is $V_i(g_\varnothing) = \frac{1}{1-\beta} w_i(g_\varnothing)$.

Like Lake and Yildiz (2016) and Lake (2018), we assume a deterministic protocol where a "leader" country (country a) has the first opportunity in each period to propose a PTA that has not yet formed. Naturally, the leader country must be a member of this PTA and the associated transition must be feasible (see Table 1). The proposed PTA forms if and only if all "recipient" countries (countries b and/or c) accept the proposal. If the leader country does not have a proposal accepted by the follower countries, one of the follower countries has the opportunity to propose a PTA. The proposal ability of the follower countries distinguishes our protocol from Aghion et al. (2007) where only the leader country can make proposals and hence, for example, the two follower countries could not form their own FTA as spokes.

Formally, Stages 1-2 describe the protocol in every period.

Stage 1(a). Country a has the opportunity to propose a PTA. If the PTA forms then the period ends. If one recipient country rejects the proposal then the game moves to Stage 1(b). If two recipient countries reject the proposal, or country a makes no proposal, then the game moves to Stage 2.[9]

Stage 1(b). Country a has the opportunity to propose a PTA with the country who did not reject its proposal in Stage 1(a). If the PTA forms then the period ends. Otherwise, the game moves to Stage 2.

Stage 2. Country b has the opportunity to propose a PTA. No matter what happens here, the period ends.

As described earlier, this protocol implies that the network remains unchanged upon the attainment of global free trade or when no agreement forms in a period. Thus, the network remains unchanged from at most the third period onwards.

As the proposer, a country can propose a PTA to which it will be a member and that has not yet formed. In our "PTA game", we allow countries to propose FTAs and CUs. Thus, Table 2 specifies the available proposals for each country as the proposer in the "PTA game" where, for example, $P_i(g)$ denotes the set of such proposals for country i and $\rho_i(g) \in P_i(g)$ denotes a proposal. More specifically, ij^{FTA} (ij^{CU}) denotes the FTA (CU) between i and j

[9] Note that, as specified in Table 1, FTA proposals involve a single recipient country. In particular, expansion from g_{ij}^{FTA} to g^{FT} proceeds via a hub-spoke network g_i^H or g_j^H. However, also as specified in Table 1, by definition, CU expansion from g_{ij}^{CU} proceeds directly to g^{FT} and hence involves two recipient countries.

	$P_i(g)$	$P_j(g)$	$P_k(g)$
$g_\varnothing$	$\{\phi, ij^{FTA}, ik^{FTA}, ij^{CU}, ik^{CU}\}$	$\{\phi, ij^{FTA}, jk^{FTA}, ij^{CU}, jk^{CU}\}$	$\{\phi, ik^{FTA}, jk^{FTA}, ik^{CU}, jk^{CU}\}$
g_{ij}^{FTA}	$\{\phi, ik^{FTA}\}$	$\{\phi, jk^{FTA}\}$	$\{\phi, ik^{FTA}, jk^{FTA}\}$
g_{ij}^{CU}	$\{\phi, ijk^{CU}\}$	$\{\phi, ijk^{CU}\}$	$\{\phi, ijk^{CU}\}$
g_i^H	$\{\phi\}$	$\{\phi, jk^{FTA}\}$	$\{\phi, jk^{FTA}\}$
g^{FT}	$\{\phi\}$	$\{\phi\}$	$\{\phi\}$

Table 2: Proposer country's action space for each subgame in PTA game

while ijk^{CU} denotes the three country CU and ϕ denotes the proposal of no PTA. In addition to our "PTA game" where countries can propose FTAs or CUs, we will also consider an "FTA game" where countries can only propose FTAs and a "CU game" where countries can only propose CUs. Naturally, as we discuss in detail later, the "FTA game" and "CU game" restrict the proposals in Table 2, respectively, by ruling out CU proposals and FTA proposals. Having received a proposal $\rho_i(g)$ from country i, each recipient country j (i.e., a country of the proposed agreement) announces a response $r_j(g, \rho_i(g)) \in \{Y, N\}$ where Y (N) denotes the acceptance (rejection) of the proposal by country j.

For each subgame at a network g, the Markov strategy of each country i must do two things: (i) specify a proposal $\rho_i(g) \in P_i(g)$ for the stage(s) where it is the proposer and (ii) assign a response $r_i\left(g, \rho_j(g)\right) \in \{Y, N\}$ to any proposal it may receive from some other country j. We solve for a type of pure strategy Markov perfect equilibrium. Specifically, using backward induction, we solve for a pure strategy subgame perfect equilibrium where the proposal by the proposer and the response by the respondent in the current period only depend on history via the network in place at the end of the previous period.[10]

Absent any further structure, the symmetry in country characteristics would generate multiple equilibria. Thus, we assume country b (c) receives an arbitrarily small non-economic benefit $\varepsilon > 0$ from bilateral PTA formation with country a rather than country c (b) and country a receives an arbitrarily small non-economic benefit $\varepsilon > 0$ from bilateral PTA formation with country b rather than country c. Indeed, these non-economic benefits can motivate our protocol ordering.[11]

Next, we use backward induction to solve for the equilibrium path of PTA networks. Section 3 analyzes the "FTA game" where countries can only form FTAs and Section 4

[10] We make two assumptions that conveniently restrict attention to certain Markov Perfect Equilibria. Given the simultaneity of responses to a proposal for CU expansion to include the CU outsider, we assume recipient countries accept such proposals if they prefer global free trade over the status quo: $r_k\left(g_{ij}^{CU}, ijk^{CU}\right) = Y$ if and only if $w_k\left(g^{FT}\right) > w_k\left(g_{ij}^{CU}\right)$. When a response $r_j(g, \rho_i(g)) = N$ merely delays formation of the PTA to a later stage of the same period, we assume the recipient country responds with $r_j(g, \rho_i(g)) = Y$. An arbitrarily small cost of making a proposal motivates this assumption.

[11] To be clear, country i's one period payoff $w_i(g)$ and continuation payoff $V_i(g)$ exclude these non-economic benefits.

	$P_i(g)$	$P_j(g)$	$P_k(g)$
$g_{\varnothing}$	$\{\phi, ij^{FTA}, ik^{FTA}\}$	$\{\phi, ij^{FTA}, jk^{FTA}\}$	$\{\phi, ik^{FTA}, jk^{FTA}\}$
g_{ij}^{FTA}	$\{\phi, ik^{FTA}\}$	$\{\phi, jk^{FTA}\}$	$\{\phi, ik^{FTA}, jk^{FTA}\}$
g_i^H	$\{\phi\}$	$\{\phi, jk^{FTA}\}$	$\{\phi, jk^{FTA}\}$
g^{FT}	$\{\phi\}$	$\{\phi\}$	$\{\phi\}$

Table 3: Proposer country's action space for each subgame in FTA game

analyzes the "CU game" where countries can only form CUs. Finally, Section 5 analyzes the "PTA game" where countries can form either FTAs or CUs.

3 Equilibrium path of networks: FTA game

In this section, we analyze the "FTA game" where countries can only form FTAs to help isolate the driving forces behind FTA formation. As specified in Table 3, our "FTA game" restricts the proposals in Table 2 by ruling out CU proposals.

Naturally, FTA formation incentives drive the equilibrium network of FTAs. In our model, these incentives include both the myopic incentives that depend on comparisons of one period payoffs across different FTA networks and the farsighted incentives that depend on comparisons of continuation payoffs across different paths of FTA networks. We now discuss the key myopic and farsighted incentives in turn.

3.1 FTA formation incentives

3.1.1 Myopic incentives

Lemma 1 describes the myopic incentives driving the equilibrium path of networks under the FTA formation game. For notational convenience, we let $\Delta w_i(g' - g) \equiv w_i(g') - w_i(g)$ and also let $g + ij^{FTA}$ denote the network that results when an FTA between countries i and j is added to network g.

Lemma 1 *(i)* $\Delta w_i(g' - g) > 0$ *for* $g' = g + ij^{FTA}$ *and* $g \neq g_{jk}^{FTA}$.

(ii) $\Delta w_i(g^{FT} - g_{jk}^{FTA}) > 0$ *for all* τ *but* $\Delta w_i(g_j^H - g_{jk}) > 0$ *if and only if* $\tau < \bar{\tau}_{OUT}^{FTA}$ *where* $\bar{\tau}_{OUT}^{FTA} \in (t_{FTA}, t_{\varnothing})$.

(iii) $\Delta w_i(g_{ij}^{FTA} - g_{jk}^{FTA}) > 0$ *for* $\tau < \tilde{\tau}$ *where* $\tilde{\tau} \in \left(\bar{\tau}_{OUT}^{FTA}, t_{\varnothing}\right)$.

(iv) $\Delta w_i(g_i^{FT} - g_{ij}^{FTA}) > 0$ *if and only if* $\tau > \bar{\tau}_{IN}^{FTA}$ *where* $\bar{\tau}_{IN}^{FTA} \in (t_{FTA}, t_{\varnothing})$.

(v) $\Delta w_i\left(g_i^H - g^{FT}\right) > 0$.

Part (i) governs the attractiveness of bilateral FTA formation. Intuitively, the preferential access exchanged through FTA formation makes bilateral FTA formation myopically attractive. However, despite the discrimination faced as an FTA outsider, the FTA outsider benefits from the lower external tariffs faced when exporting to the FTA insiders that result from FTA insiders practicing tariff complementarity. Thus, the one exception to countries finding bilateral FTA formation attractive is that an FTA outsider *might* not benefit from a bilateral FTA.

Parts (ii)-(iii) describe the FTA outsider's "free riding" incentives. On one hand, through eliminating the discrimination faced in both FTA insider markets, an FTA outsider always benefits from global free trade. On the other hand, an FTA outsider only benefits from becoming the spoke when the tariff binding is sufficiently low. When the tariff binding does not bind the FTA outsider, i.e. $\tau > t_{\varnothing}$, the FTA outsider benefits from the tariff complementarity practiced by FTA insiders and from imposing its optimal tariff on both FTA insiders. Indeed, these benefits are large enough that a country prefers being an FTA outsider than an FTA insider. However, the tariff binding constrains this ability to impose optimal tariffs on the FTA insiders once $\tau < t_{\varnothing}$. Once the tariff binding falls below $\tilde{\tau}$, part (iii) says a country prefers being an FTA insider over an FTA outsider. But, as long as the tariff binding does not bind too tightly, i.e. $\tau > \bar{\tau}_{OUT}^{FTA}$, part (ii) says the FTA outsider still prefers remaining an FTA outsider over becoming a spoke. Here, we say the FTA outsider has a *myopic free riding incentive*. Later, the tension for an FTA outsider between the myopic free riding incentive and the farsighted incentive for global free trade creates a dynamic free riding incentive.

The impact of the falling tariff binding on the FTA outsider's ability to impose its optimal tariff also alters the incentive of FTA insiders to exclude the FTA outsider. When the tariff binding is sufficiently high, the FTA outsider imposes relatively high tariffs on the FTA insiders. To eliminate these barriers, the FTA insiders have a strong incentive for FTA expansion to global free trade. However, the strength of this incentive weakens as the FTA outsider becomes more constrained by a tighter tariff binding. Indeed, part (iii) says that FTA insiders have incentives to exclude the FTA outsider from FTA expansion to global free trade once the tariff binding is sufficiently tight. In this case, we say the FTA insiders have an *exclusion incentive*. Intuitively, the relatively tight tariff binding on the FTA outsider allows FTA insiders to extract tariff concessions from the FTA outsider without engaging in FTA formation.

Part (iv) represents the "flexibility benefit" of FTAs. Because of the sovereign discretion over external tariffs, each FTA insider has the flexibility to form its own subsequent FTA and become the hub. This FTA flexibility benefit is valuable, i.e. $w_i \left(g_i^H - g^{FT}\right) > 0$, because the

hub country has preferential access to each spoke market that is eroded by FTA formation between the spokes that yields global free trade.

We now move on to discuss the farsighted incentives that drive FTA formation.

3.1.2 Farsighted incentives

Our equilibrium analysis relies on two farsighted incentives. Lemma 2 presents these incentives.

Lemma 2 *(i)* $V_z\left(g_{ij}^{FTA}\right) > V_z\left(g_{\varnothing}\right)$ *for* $z = i, j$.

(ii) $V_i\left(g_{ij}^{FTA}\right) > V_i\left(g_{jk}^{FTA}\right)$ *when* $g_{ij}^{FTA} \to g_i^H \to g^{FT}$ *and* $g_{jk}^{FTA} \to g_j^H \to g^{FT}$.

Part (i) of Lemma 2 imposes a participation constraint for FTA insiders. Given Lemma 1(i), these participation constraints hold if FTA formation does not expand beyond a single FTA. Thus, the participation constraints really pertain to situations where FTA formation expands beyond a single FTA and, given spokes form their own FTA by Lemma 1(i), where such expansion proceeds to global free trade. In turn, the participation constraints for FTA insiders say that their continuation payoff from a path of FTAs leading to global free trade exceeds their continuation payoff in a world without any PTAs. Given the benefit of being the hub, the tightest participation constraint is for the FTA insider-turned-spoke. In particular, an FTA insider prefers being a insider-turned-spoke on the path to global free trade over the world without any PTAs.

Given the benefits of tariff complementarity for an FTA outsider, it is not necessarily true that a country myopically prefers being an FTA insider over an FTA outsider. Nevertheless, part (ii) says that, when FTA formation expands to global free trade, a country's continuation payoff as an FTA insider-turned-hub exceeds that as an FTA outsider-turned-spoke. Intuitively, if a country happily participates in FTA expansion to global free trade as an FTA outsider then the myopic free riding incentive is sufficiently weak relative that the benefit of being the hub ensures it prefers being the FTA insider-turned-hub over the FTA outsider-turned-spoke.[12]

We now move on and solve the equilibrium path of networks in the FTA game by backward induction.

[12] Note that whether a country prefers being an FTA insider-turned-spoke or an FTA outsider-turned-spoke on the path to global free trade merely depends on whether a country myopically prefers being an FTA insider or an FTA outsider. That is, when $g_{ij}^{FTA} \to g_j^H \to g^{FT}$ and $g_{jk}^{FTA} \to g_j^H \to g^{FT}$ then $V_i\left(g_{ij}^{FTA}\right) > V_i\left(g_{jk}^{FTA}\right)$ if and only if $w_i\left(g_{ij}^{FTA}\right) > w_i\left(g_{jk}^{FTA}\right)$.

3.2 Subgames at hub-spoke networks

To begin the backward induction, consider a subgame at a hub-spoke network g_i^H. The following lemma follows directly from Lemma 1(i):

Lemma 3 *In the subgame at a hub-spoke network g_i^H, spoke countries form the final FTA that leads to global free trade: $g_i^H \rightarrow g^{FT}$.*

3.3 Subgames at FTA insider-outsider networks

We now roll back to a subgame at an insider-outsider network g_{ij}^{FTA}. In general, tensions between myopic and farsighted FTA formation incentives exist for both the FTA outsider and FTA insiders. We first explore this tension facing the FTA outsider.

3.3.1 Dynamic free riding incentive

A tension between myopic and farsighted incentives for the FTA outsider creates the possibility of a dynamic free riding incentive. As explained with Lemma 1(ii), by eliminating the discrimination faced in FTA insider markets, an FTA outsider always benefits from moving directly to global free trade. Thus, given spokes form the final FTA leading to global free trade, an FTA outsider has a farsighted incentive to become a spoke and subsequently reach global free trade: $\Delta w_k(g^{FT} - g_{ij}^{FTA}) > 0$. However, an FTA outsider has a myopic incentive to refuse subsequent FTA formation when it holds a myopic free riding incentive. As described in Lemma 1(ii), the presence of a myopic free riding incentive depends on the tariff binding: $\Delta w_k(g_i^H - g_{ij}^{FTA}) > 0$ if and only if $\tau < \bar{\tau}_{OUT}^{FTA}$. That is, an FTA outsider does not hold a myopic free riding incentive when the tariff binding is sufficiently tight because such a tariff binding severely constrains its ability to impose optimal tariffs on the FTA insiders. In this case, there is no dynamic free riding incentive because an FTA outsider happily participates in subsequent FTA formation from myopic and farsighted perspectives.

However, the FTA outsider holds a myopic free riding incentive with a sufficiently high tariff binding $\tau > \bar{\tau}_{OUT}^{FTA}$. Here, the FTA outsider benefits from the tariff complementarity practiced by FTA insiders and faces little constraint on its optimal tariffs. In turn, the discount factor mediates the farsighted incentive of the FTA outsider to become a spoke and its myopic free riding incentive. Specifically, the FTA outsider prefers becoming a spoke rather than remaining a permanent FTA outsider when

$$w_k\left(g_i^H\right) + \frac{\beta}{1-\beta} w_k\left(g^{FT}\right) > \frac{1}{1-\beta} w_k\left(g_{ij}^{FTA}\right) \tag{16}$$

which, as one would expect from our above discussion, can only fail if the FTA outsider holds a myopic free riding incentive $\Delta w_k(g_{ij}^{FTA} - g_i^H) > 0$. (16) reduces to

$$\beta > \bar{\beta}_{OUT}(\tau) \equiv \left[1 + \frac{\Delta w_k \left(g^{FT} - g_{ij}^{FTA}\right)}{\Delta w_k \left(g_{ij}^{FTA} - g_i^H\right)}\right]^{-1} \tag{17}$$

which says a sufficiently patient FTA outsider becomes a spoke or, alternatively, a sufficiently myopic FTA outsider refuses subsequent FTA formation. In this latter case, i.e. $\beta < \bar{\beta}_{OUT}(\tau)$, we say the FTA outsider has a dynamic free riding incentive. Moreover, the extent that the FTA outsider holds a dynamic free riding incentive rises (i.e. $\bar{\beta}_{OUT}(\tau)$ falls) as the myopic free riding incentive, $\Delta w_k \left(g_{ij}^{FTA} - g_i^H\right)$ increases relative to the farsighted incentive of becoming a spoke $\Delta w_k \left(g^{FT} - g_{ij}^{FTA}\right)$.

Figure 3 illustrates the dynamic free riding incentive and how it varies with the tariff binding. An FTA outsider has a dynamic free riding incentive when $\beta < \bar{\beta}_{OUT}(\tau)$. In this case, the FTA outsider is sufficiently myopic that the myopic free riding incentive dominates the farsighted incentive to become a spoke and, hence, the FTA outsider refuses subsequent FTA formation. In the "no binding region" of $\tau > t_{\varnothing}$, the FTA outsider's optimal applied tariff remains unbound and, in turn, stays unchanged. Thus, $\bar{\beta}_{OUT}(\tau)$ remains constant. However, the FTA outsider's optimal tariff becomes bound once we move into the "non-member binding region" where $\tau < t_{\varnothing}$. By reducing the FTA outsider's ability to impose optimal tariffs on FTA insiders, the myopic free riding incentive weakens and the farsighted incentive to become a spoke strengthens. In turn, the dynamic free riding incentive weakens, i.e. $\bar{\beta}_{OUT}(\tau)$ falls, as the tariff binding continues falling below $t_{\varnothing}$. Indeed, the dynamic free riding incentive disappears once the tariff binding falls below $\bar{\tau}_{OUT}^{FTA}$ ($\bar{\beta}_{OUT}(\tau)$ becomes negative) because the myopic free riding incentive disappears and, hence, regardless of the discount factor, the FTA outsider becomes a spoke.

Next, we examine the myopic and farsighted incentives of FTA insiders.

3.3.2 Dynamic exclusion incentive

A tension between myopic and farsighted incentives for FTA insiders create the possibility of a dynamic exclusion incentive. Myopically, having sole preferential access in both spoke markets makes FTA formation with the FTA outsider attractive for an FTA insider: $\Delta w_i(g_i^H - g_{ij}^{FTA}) > 0$. However, an FTA insider knows that the spokes will then form their own FTA and precipitate global free trade which erodes the preferential access enjoyed with its FTA insider partner. Indeed, if an FTA insider holds an exclusion incentive if it benefits from permanently excluding the FTA outsider from subsequent FTA formation and such an exclusion incentive provides a farsighted incentive to refrain from becoming the hub. As de-

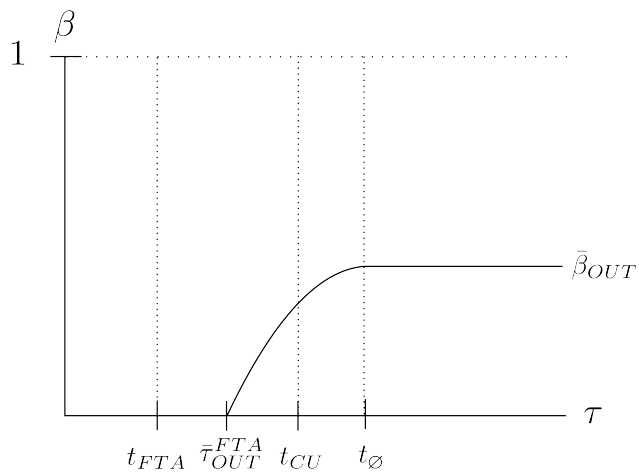

Figure 3: Dynamic free riding incentive

scribed in Lemma 1(iii), the presence of an exclusion incentive depends on the tariff binding: $\Delta w_i(g_{ij}^{FTA} - g^{FT}) > 0$ if and only if $\tau < \bar{\tau}_{IN}^{FTA}$. That is, an FTA insider does not hold an exclusion incentive when the binding is sufficiently lax because the FTA outsider still levies relatively high tariffs which makes FTA formation with the FTA outsider quite attractive. In this case, there is no dynamic exclusion incentive because FTA formation with the FTA outsider is attractive from both myopic and farsighted perspectives.

However, an FTA insider holds an exclusion incentive with a sufficiently tight tariff binding. Here, by constraining the FTA outsider's applied tariff, the FTA insiders extract substantial tariff concessions from the FTA outsider without any reciprocal concessions to the FTA outsider. In turn, the discount factor mediates the myopic incentive to become the hub and the farsighted incentive to exclude the FTA outsider. Specifically, an FTA insider prefers to become the hub rather than remain a permanent FTA insider when

$$w_i\left(g_i^H\right) + \frac{\beta}{1-\beta} w_i\left(g^{FT}\right) > \frac{1}{1-\beta} w_i\left(g_{ij}^{FTA}\right). \tag{18}$$

which, as one would expect from our above discussion, can only fail if the FTA insider holds an exclusion incentive $\Delta w_i(g_{ij}^{FTA} - g^{FT}) > 0$. (18) reduces to

$$\beta < \bar{\beta}_{IN}(\tau) \equiv \left[1 + \frac{\Delta w_i\left(g_{ij}^{FTA} - g^{FT}\right)}{\Delta w_i\left(g_i^H - g_{ij}^{FT}\right)}\right]^{-1}. \tag{19}$$

which says a sufficiently myopic FTA insider becomes the hub or, alternatively, a sufficiently patient FTA insider refuses subsequent FTA formation with the FTA outsider. In this latter case, i.e. $\beta > \bar{\beta}_{IN}(\tau)$, we say the FTA insider has a dynamic exclusion incentive. Moreover, the extent that the FTA insider holds a dynamic exclusion incentive rises (i.e. $\bar{\beta}_{IN}(\tau)$ falls)

as the exclusion incentive, $\Delta w_i \left(g_{ij}^{FTA} - g_i^{FT}\right)$ increases relative to the myopic incentive of becoming the hub $\Delta w_i \left(g_i^H - g_{ij}^{FTA}\right)$.

Figure 4 illustrates the dynamic exclusion incentive and how it varies with the tariff binding τ. An FTA insider has a dynamic exclusion incentive when $\beta > \bar{\beta}_{IN}(\tau)$. In this case, an FTA insider is sufficiently patient that the farsighted nature of the exclusion incentive dominates the myopic incentive to become the hub and, hence, the FTA insider refuses subsequent FTA formation with the FTA outsider. Naturally, a pre-requisite for the dynamic exclusion incentive, i.e. $\bar{\beta}_{IN}(\tau) < 1$, is that the FTA insider actually holds an exclusion incentive. Further, as discussed above, $\Delta w_i \left(g_{ij}^{FTA} - g^{FT}\right) > 0$ requires sufficiently tight tariff bindings $\tau < \bar{\tau}_{IN}^{FTA}$ so that the FTA insider extracts substantial tariff concessions from the FTA outsider without engaging in FTA formation with the FTA outsider. As Figure 4 shows, the dynamic exclusion continues strengthening, i.e. $\bar{\beta}_{IN}(\tau)$ falls, as the tariff binding falls through FTA binding region $\tau < t_{FTA}$.

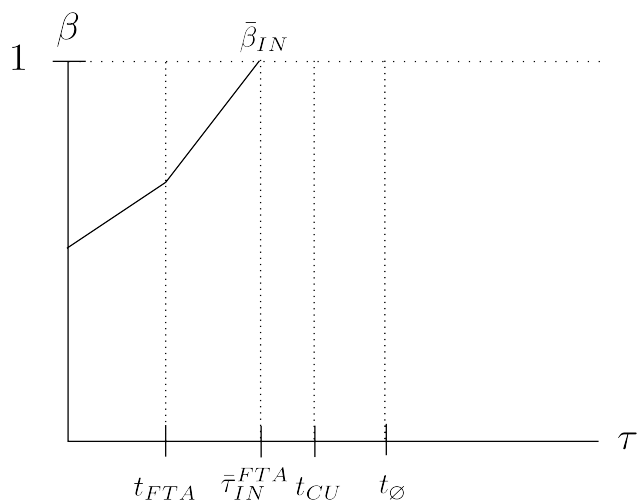

Figure 4: Dynamic exclusion incentive

The dynamic free riding incentive faced by the FTA outsider and the dynamic exclusion incentive faced by the FTA insiders interact in determining whether subsequent FTA formation takes place. Put simply, subsequent FTA formation takes place between an FTA insider and the FTA outsider if and only if the FTA outsider does not hold a dynamic free riding incentive and the FTA insiders do not hold a dynamic exclusion incentive. Otherwise, the FTA outsider exploits its dynamic free riding incentive and free rides on FTA formation by the FTA insiders or the FTA insiders exploit their dynamic exclusion incentive and exclude the FTA outsider from subsequent FTA formation. Lemma 4 summarizes these findings.

Lemma 4 *Consider a subgame at an FTA insider-outsider network g_{ij}^{FTA} where country i proposes before country j. Then, the equilibrium outcome in the subgame is*

(i) An FTA between the outsider and an insider (i.e. $g_{ij}^{FTA} \to g_i^H$*) if* $\beta \in \left(\bar{\beta}_{IN}(\tau), \bar{\beta}_{OUT}(\tau)\right)$
(ii) No FTA (i.e. $g_{ij}^{FTA} \to g_{ij}^{FTA}$*) if* $\beta \notin \left(\bar{\beta}_{IN}(\tau), \bar{\beta}_{OUT}(\tau)\right)$.

3.4 Subgame at empty network

Rolling back to the subgame at the empty network $g_\varnothing$ and solving the equilibrium outcome in this subgame reveals the equilibrium path of networks. Moreover, the intuition discussed above regarding the dynamic free riding incentive of the FTA outsider and the dynamic exclusion incentive of FTA insiders, and how these vary with the tariff binding, drives the equilibrium path of networks.

Proposition 1 summarizes the equilibrium path of networks.

Proposition 1 *The equilibrium path of networks in the FTA game is*
(i) $g_\varnothing \to g_{ac}^{FTA} \to g_a^H \to g^{FT}$ *when* $\beta \in \left(\bar{\beta}_{IN}(\tau), \bar{\beta}_{OUT}(\tau)\right)$ *and* $\tau \geq \tilde{\tau}$
(ii) $g_\varnothing \to g_{ab}^{FTA} \to g_a^H \to g^{FT}$ *when* $\beta \in \left(\bar{\beta}_{IN}(\tau), \bar{\beta}_{OUT}(\tau)\right)$ *and* $\tau < \tilde{\tau}$
(iii) $g_\varnothing \to g_{ab}^{FTA}$ *when* $\beta \notin \left(\bar{\beta}_{IN}(\tau), \bar{\beta}_{OUT}(\tau)\right)$.

Proposition 1 says the equilibrium path of networks revolves around the dynamic free riding incentive of an FTA outsider and the dynamic exclusion incentive of an FTA insider. If neither the FTA outsider holds a dynamic exclusion incentive (i.e. $\beta > \bar{\beta}_{OUT}(\tau)$) nor an FTA insider holds a dynamic exclusion incentive (i.e. $\beta < \bar{\beta}_{IN}(\tau)$), the leader country, country a, becomes the hub on a path of FTAs leading to global free trade. However, the leader country is a member of a permanent FTA if either the FTA outsider holds a dynamic free riding incentive or the FTA insiders hold a dynamic exclusion incentive (i.e. $\beta \notin \left(\bar{\beta}_{IN}(\tau), \bar{\beta}_{OUT}(\tau)\right)$). Thus, the dynamic free riding incentive and the dynamic exclusion incentive drive the equilibrium.

One subtlety emerges regarding the equilibrium path of networks. When $\tau \geq \tilde{\tau}$, the tariff binding is so lax that the myopic free riding incentive is sufficiently strong that countries myopically prefer being an FTA outsider than an FTA insider. Thus, when FTA formation expands to global free trade, country b would prefer being an FTA outsider-turned spoke over an FTA insider-turned-spoke on the path to global free trade. However, if country b were to make an FTA proposal in stage 2 of the protocol, the attractiveness of being the hub mean it would propose an FTA with country c and be the FTA insider-turned-hub on the path to global free trade. This threat implies country b can credibly reject an FTA proposal from country a (in stage 1a) knowing that such rejection will force country a to propose FTA formation with country c (in stage 1b) and leave country b as the FTA outsider-turned-spoke. However, when FTA formation does not expand to global free trade,

the non-economic benefits imply country b will propose FTA formation with country a if it were to make an FTA proposal in stage 2. However, now country b cannot credibly reject an FTA proposal from country a in stage 1a because it knows it will eventually accept such a proposal.

Figure 5 shows how tariff binding liberalization changes the incentives that constrain FTA formation from reaching global free trade. For sufficiently high tariff bindings $\tau > \bar{\tau}_{IN}^{FTA}$, FTA insiders do not hold an exclusion incentive (i.e. $\Delta w_i \left(g^{FT} - g_{ij}^{FTA}\right) > 0$) because the relatively high applied tariffs of the FTA outsider create strong incentives for FTA formation with the FTA outsider. In turn, a dynamic exclusion incentive does not exist. However, an FTA outsider holds a myopic free riding incentive whereby subsequent FTA formation is myopically unattractive. The relatively lax tariff bindings ($\tau > \bar{\tau}_{OUT}^{FTA}$) imply the FTA outsider would give relatively large tariff concessions via FTA formation but receive relatively small concessions due to the tariff complementarity practiced by FTA insiders. In turn, a sufficiently myopic FTA outsider holds a dynamic free riding incentive and thus refuses subsequent FTA formation with an FTA insider. Ultimately, with relatively high tariff bindings, the dynamic free riding incentive constrains the attainment of global free trade.

However, as tariff bindings continually fall, the dynamic free riding incentive eventually disappears and the dynamic exclusion incentive becomes the force constraining the attainment of global free trade. Once the tariff binding falls below $\bar{\tau}_{OUT}^{FTA}$, the tariff binding constrains the FTA outsider such that the tariffs it imposes on the FTA insiders differ little from those faced by the FTA outsider. This makes the discrimination faced by the FTA outsider more prominent and eliminates any dynamic free riding incentive. However, the relatively low tariff binding means the FTA insiders have already extracted substantial tariff concessions from the FTA outsider and this generates an exclusion incentive for FTA insiders. In turn, despite the myopic incentive to become the hub, a sufficiently patient FTA insider holds a dynamic exclusion incentive and thus refuses subsequent FTA formation with the FTA outsider. Ultimately, with relatively low tariff bindings, the dynamic exclusion incentive of FTA insiders constrains the attainment of global free trade.

With an understanding of the forces constraining the extent of FTA formation and how these forces vary with the tariff binding, we now ask whether multilateral trade liberalization through lower tariff bindings can expand the extent of FTA formation. Specifically, when FTA formation stops short of global free trade under high tariff bindings, can lower tariff bindings propel FTA formation to global free trade? Corollary 1 answers this question.

Corollary 1 *Consider sufficiently high tariff bindings $\tau > \bar{\tau}_{IN}^{FTA}$. Then:*

(i) When global free trade is not attained for $\tau > \bar{\tau}_{IN}^{FTA}$, multilateral tariff binding liberalization facilitates FTA formation in achieving global free trade.

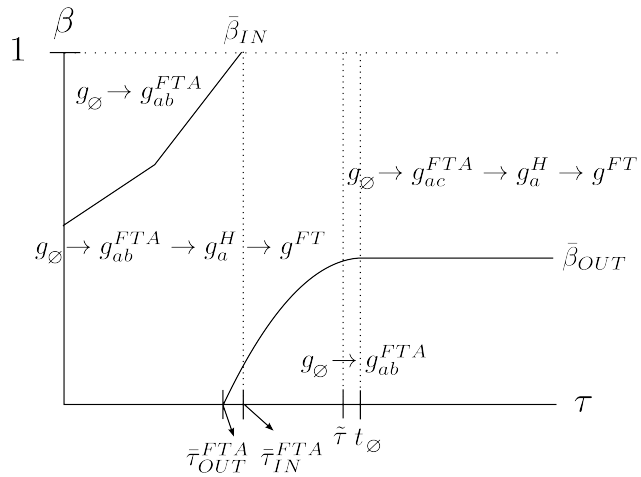

Figure 5: Equilibrium path of networks: FTA game

(ii) When global free trade is attained for $\tau > \bar{\tau}_{IN}^{FTA}$, multilateral tariff binding liberalization never facilitates but can impede FTA formation in achieving global free trade.

Corollary 1 highlights the fundamental importance of multilateral tariff binding liberalization for the prospect of FTA expansion leading to global free trade. With sufficiently high tariff bindings, the dynamic free riding incentive constrains the attainment of global free trade. In particular, when global free trade is not attained for $\tau > \bar{\tau}_{IN}^{FTA}$ then countries are sufficiently myopic, $\beta < \bar{\beta}_{OUT}(t_{\varnothing})$, that the myopic free riding incentive can generate the dynamic free riding incentive. Specifically, for any tariff binding $\tau > \bar{\tau}_{IN}^{FTA}$, the FTA outsider has a dynamic free riding incentive when $\beta < \bar{\beta}_{OUT}(\tau)$ and this prevents FTA expansion to global free trade. Nevertheless, tariff binding liberalization weakens the dynamic free riding incentive by constraining the FTA outsider's ability to impose tariffs on the FTA insiders. Formally, this raises the extent that global free trade is attained by reducing $\bar{\beta}_{OUT}(\tau)$ and implies that multilateral tariff binding liberalization complements FTA formation in attaining global free trade.

Conversely, the dynamic exclusion incentive constrains the attainment of global free trade with sufficiently low tariff bindings and further multilateral tariff binding liberalization further constrains the attainment of global free trade. In particular, when global free trade is attained for $\tau > \bar{\tau}_{IN}^{FTA}$ then countries are sufficiently patient $\beta > \bar{\beta}_{IN}(\tau \approx 0)$ and do not hold a dynamic exclusion incentive nor a dynamic free riding incentive. However, for any tariff binding $\tau < \bar{\tau}_{OUT}^{FTA}$, the FTA insiders have a dynamic exclusion incentive when they are sufficiently patient, $\beta > \bar{\beta}_{IN}(\tau)$, and this prevents FTA expansion to global free trade. Moreover, further liberalization only increases the magnitude of tariff concessions that FTA insiders extract from the FTA outsider without having to form FTAs with the outsider. This

	$P_i(g)$	$P_j(g)$	$P_k(g)$
$g_{\varnothing}$	$\{\phi, ij^{CU}, ik^{CU}\}$	$\{\phi, ij^{CU}, jk^{CU}\}$	$\{\phi, ik^{CU}, jk^{CU}\}$
g_{ij}^{CU}	$\{\phi, ijk^{FTA}\}$	$\{\phi, ijk^{CU}\}$	$\{\phi, ijk^{CU}\}$
g^{FT}	$\{\phi\}$	$\{\phi\}$	$\{\phi\}$

Table 4: Proposer country's action space for each subgame in CU game

strengthens the dynamic exclusion incentive. Formally, multilateral tariff binding liberalization reduces the extent that global free trade is attained by reducing $\bar{\beta}_{IN}(\tau)$ and implies that multilateral tariff binding liberalization impedes FTA formation in achieving global free trade.[13]

4 Equilibrium path of networks: CU game

In this section, we analyze the "CU game" where countries can only form CUs. Analogous to our "FTA game" that restricted the proposals in Table 2 by ruling out CU proposals, Table 4 shows how our "CU game" restricts the proposals in Table 2 by ruling out FTA proposals.

The different tariff setting behavior of FTA insiders versus CU insiders has an important implication for the ability of PTA members to form subsequent PTAs. Specifically, because CU insiders impose a common external tariffs on non-members, they cannot individually form their own subsequent PTA. Rather, CU insiders can only engage in subsequent PTA formation jointly with, and with the consent of, its CU insider partner. As we will see in Section 5 that analyzes the "PTA game" where countries can form FTAs or CUs, this has important implications on the PTA member and PTA non-member incentives.

4.1 CU formation incentives

In contrast to FTA formation, CU formation does not generate a tension between myopic and farsighted incentives. Underlying this observation is that CU expansion must move directly from the CU insider-outsider network to global free trade and CU insiders have veto power over such CU expansion. Lemma 5 describes the myopic incentives driving the equilibrium path of networks in the CU game.

Lemma 5 *(i)* $\Delta w_i(g_{ij}^{CU} - g_{\varnothing}) > 0$ *but* $\Delta w_k(g_{ij}^{CU} - g_{\varnothing}) < 0$

[13] When β lies in the intermediate range $\beta \in \left(\bar{\beta}_{OUT}(\tau = t_{\varnothing}), \bar{\beta}_{IN}(\tau = 0)\right)$, Figure 5 shows that tariff binding liberalization neither facilitates nor impedes FTA formation in achieveing global free trade.

(ii) $\Delta w_k(g^{FT} - g_{ij}^{CU}) > 0$ *for all* τ *but* $\Delta w_i(g^{FT} - g_{ij}^{CU}) > 0$ *if and only if* $\tau > \bar{\tau}_{IN}^{CU}$ *where* $\bar{\tau}_{IN}^{CU} \in (t_{CU}, t_{\varnothing})$.

(iii) $\Delta w_i(g_{ij}^{CU} - g_{ij}^{FTA}) > 0$ *for* $\tau > t_{FTA}$.

Part (i) of Lemma 5 governs the impact of CU formation on CU insiders and the CU outsider. First, CU insiders benefit from CU formation which is not surprising given that FTA insiders benefit from FTA formation and the joint determination of external tariffs allows CU insiders to internalize the negative intra-PTA externality of tariff complementarity. Second, unlike the outsider who could benefit from FTA formation, the CU outsider always suffers from CU formation. While CU insiders may practice tariff complementarity, their internalization of the negative intra-PTA externality of tariff complementarity implies the discrimination faced by the CU outsider always dominates the benefit of any tariff complementarity.

Part (ii) of Lemma 5 governs the incentives faced by CU insiders and the CU outsider for CU expansion to global free trade. First, the CU outsider always benefits from CU expansion to global free trade. This is not surprising given that the FTA outsider always benefited from simultaneously eliminating the discrimination faced in *both* FTA insider markets and that the CU outsider faces a greater degree of discrimination than the FTA outsider. Thus, the CU outsider does not hold any type of free riding incentive. Second, similar to FTA insiders, CU insiders have an incentive to exclude the CU outsider from global free trade under sufficiently tight tariff bindings: $\Delta w_i\left(g^{FT} - g_{ij}^{CU}\right) < 0$, when $\tau < \bar{\tau}_{IN}^{CU}$. Indeed, given CU insiders internalize the negative intra-PTA externality of tariff complementarity, this CU exclusion incentive is stronger than the FTA exclusion incentive so that $\bar{\tau}_{IN}^{FTA} < \bar{\tau}_{IN}^{CU}$.

Although not relevant for our CU game, part (iii) captures an important incentive for the PTA game. As discussed above, CU insiders internalize the negative intra-PTA externality of tariff complementarity through coordinating their external tariffs. In contrast, by setting individually optimal external tariffs, FTA insiders impose a negative externality on each other through tariff complementarity. Thus, part (iii) describes the myopic coordination benefit of CU formation. Naturally, this benefit disappears once the tariff binding actually binds FTA and CU insiders because then they both set the same external tariffs.

Ultimately, the fact that CU expansion proceeds directly from the CU insider-outsider network to global free trade, unlike FTA formation which proceeded through the hub-spoke network, implies there is no tension between myopic and farsighted CU formation incentives. Moreover, given the CU outsider always benefits from CU expansion to global free trade, the CU exclusion incentive of CU insiders is the incentive that constrains the attainment of global free trade. Thus, CU expansion takes place if and only if CU insiders do not hold an exclusion incentive with the leader country always proposing CU formation and being a CU

insider. Proposition 2 summarizes this result and Figure 6 illustrates.

Proposition 2 *The equilibrium path of networks in the CU game is*
(i) $g_{\varnothing} \longrightarrow g_{ab}^{CU} \longrightarrow g^{FT}$ *when* $\tau > \bar{\tau}_{IN}^{CU}$
(ii) $g_{\varnothing} \rightarrow g_{ab}^{CU}$ *when* $\tau < \bar{\tau}_{IN}^{CU}$.

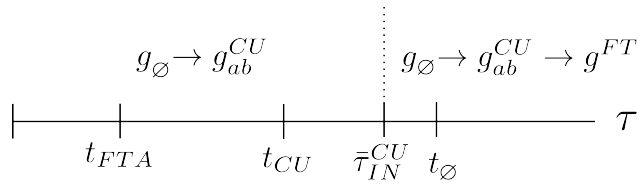

Figure 6: Equilibrium path of networks: CU game

So far, we have examined the FTA and CU games in isolation. Next, we endogenize the choice between FTAs and CUs.

5 Equilibrium path of networks: PTA game

Unlike earlier sections that allowed countries to only form FTAs (the "FTA game") or only form CUs (the "CU game"), we now allow countries to choose between forming FTAs or CUs. That is, we endogenously determine the choice between FTAs and CUs along the equilibrium path of networks. As we explain below, the key trade-off introduced by the endogenous choice between FTAs and CUs is a trade-off between the flexibility benefit of FTAs and the coordination benefits of CUs.

5.1 FTA Flexibility benefit

As stated before, the central difference between an FTA and a CU is that members of a CU impose common external tariffs on non-members whereas FTA members adopt individually optimal tariffs. This different tariff setting behavior has an important implication for subsequent PTA formation: while an FTA member can freely form additional FTAs with non-member countries without the consent of existing FTA partners, CU members must form additional CUs together and any initial CU member has veto power. That is, unlike CU members who must jointly form a subsequent PTA with the CU outsider, FTA members have the flexibility to form their own individual FTAs with the FTA outsider.

The flexibility of FTAs confers an FTA flexibility benefit on FTA insiders. Unlike CU insiders who lose preferential access with each other upon CU expansion to global free trade, an FTA insider enjoys sole preferential access to both spoke markets upon becoming the

hub via FTA expansion. Formally, Lemma 1(iv) captured this FTA flexibility benefit as $w_i\left(g_i^H - g^{FT}\right) > 0$. That is, the FTA flexibility benefit for an FTA insider captures the higher payoff from becoming the hub via subsequent FTA formation relative to subsequent CU formation which must proceed directly to global free trade.

5.2 Myopic and farsighted CU coordination benefits

Unlike FTA members who retain sovereign discretion over their external tariffs, CU insiders coordinate their external tariffs and this coordination generates myopic and farsighted coordination benefits. Myopically, coordination of external tariffs allows internalization of the negative intra-PTA externality posed by tariff complementarity. As long as the tariff binding lies above the "FTA binding" region, i.e. $\tau > t_{FTA}$, the tariff binding does not completely eliminate the CU insiders' ability to internalize this externality: $w_i\left(g_{ij}^{CU} - g_{ij}^{FTA}\right) > 0$ if $\tau > t_{FTA}$.

Moreover, the fact that CU expansion requires joint approval of CU insiders, due to the requirement of common external tariff, can generate a farsighted CU coordination benefit. Specifically, while each CU insider can *veto* CU expansion to global free trade, each FTA insider can *precipitate* global free trade by exploiting the FTA flexibility benefit to become the hub on the path to global free trade. Thus, $w_i\left(g_{ij}^{CU} - g^{FT}\right) > 0$ not only represents the CU exclusion incentive but and also the farsighted coordination benefit of CU formation in that PTA insiders can form CUs and, due to external tariff coordination, ensure PTA expansion does not take place. As discussed above, CU insiders hold a CU exclusion incentive, and hence CU formation has a farsighted coordination benefit, when the tariff binding is sufficiently tight, $\tau < \bar{\tau}_{IN}^{CU}$.

5.3 Trade-off between FTA flexibility benefit and CU coordination benefits

The trade off between the FTA flexibility benefit and CU coordination benefits generates a dynamic trade off. Further, the exact nature of the trade-off depends on whether CU insiders hold an exclusion incentive.

In general, country i prefers being an FTA insider and then the hub on the path to global free trade over being a CU insider when

$$V_i\left(g_{ij}^{FTA}\right) = w_i\left(g_{ij}^{FTA}\right) + \beta w_i\left(g_i^H\right) + \frac{\beta^2}{1-\beta} w_i\left(g^{FT}\right) > V_i\left(g_{ij}^{CU}\right). \tag{20}$$

When CU insiders do not hold a CU exclusion incentive, and hence there is no farsighted

CU coordination benefit, CU formation expands to global free trade. Then, $V_i\left(g_{ij}^{CU}\right) = w_i\left(g_{ij}^{CU}\right) + \frac{\beta}{1-\beta} w_i\left(g^{FT}\right)$ and we can rewrite (20) as

$$\beta \underbrace{\left[w_i\left(g_i^H\right) - w_i\left(g^{FT}\right)\right]}_{\text{FTA flexibility benefit}} > \underbrace{\left[w_i\left(g_{ij}^{CU}\right) - w_i\left(g_{ij}^{FTA}\right)\right]}_{\text{myopic CU coordination benefit}}. \tag{21}$$

That is, a country prefers being an FTA insider and then the hub on the path to global free trade over a CU insider on the path to global free trade if and only if the discounted FTA flexibility benefit dominates the myopic CU coordination benefit. In terms of the threshold discount factor, (23) reduces to

$$\beta > \underline{\beta}^{Flex}(\tau) \equiv \frac{w_i\left(g_{ij}^{CU} - g_{ij}^{FTA}\right)}{w_i\left(g_i^H - g^{FT}\right)} \tag{22}$$

where $\underline{\beta}_{Flex}(\tau)$ measures the size of the myopic CU coordination benefit relative to the FTA flexibility benefit. As the FTA flexibility benefit grows relative to myopic CU coordination benefit then $\underline{\beta}_{Flex}(\tau)$ falls and, thus, the extent of FTA formation expands.

However, when CU insiders hold a CU exclusion incentive, i.e. $\tau < \bar{\tau}_{IN}^{CU}$, they hold a farsighted CU coordination benefit and this modifies the flexibility-coordination trade off. Given CU insiders exclude the CU outsider from CU expansion, $V_i\left(g_{ij}^{CU}\right) = \frac{1}{1-\beta} w_i\left(g_{ij}^{CU}\right)$ and we can rewrite (20) as

$$\beta \underbrace{\left[w_i\left(g_i^H\right) - w_i\left(g^{FT}\right)\right]}_{\text{FTA flexibility benefit}} > \underbrace{\left[w_i\left(g_{ij}^{CU}\right) - w_i\left(g_{ij}^{FTA}\right)\right]}_{\text{myopic CU coordination benefit}} + \frac{\beta}{1-\beta} \underbrace{\left[w_i\left(g_{ij}^{CU}\right) - w_i\left(g^{FT}\right)\right]}_{\text{farsighted CU coordination benefit}}. \tag{23}$$

That is, a country prefers being an FTA insider and then the hub on the path to global free trade over a permanent CU insider if and only if the discounted FTA flexibility benefit dominates the myopic CU coordination benefit and the discounted farsighted CU coordination benefit. In terms of the threshold discount factor, (23) reduces to

$$\beta \in \left(\underline{\beta}^{Flex}(\tau), \bar{\beta}^{Flex}(\tau)\right). \tag{24}$$

Again, a stronger FTA flexibility relaxes the inequality in (23) and expands the extent of FTA formation while a stronger myopic CU coordination benefit tightens the inequality in (23) and constrains the extent of FTA formation. However, now the farsighted part of the CU coordination benefit, as captured by the size of the CU exclusion incentive $w_i\left(g_{ij}^{CU} - g^{FT}\right) > 0$, also tightens the inequality in (23). Thus, as either the myopic or farsighted part of the CU coordination benefit become stronger then the extent of FTA formation falls.

Figure 7 illustrates how the trade off between the FTA flexibility benefit and CU coordination benefits change with the tariff binding. In general, the CU coordination benefits consist of a myopic component, revolving around internalization of tariff complementarity by CU insiders, and a farsighted component, revolving around a desire by CU insiders to exclude the CU outsider. However, CU insiders do not hold an exclusion incentive when the tariff binding exceeds $\bar{\tau}_{IN}^{CU}$. Thus, for $\tau > \bar{\tau}_{IN}^{CU}$, the CU coordination benefit is merely the myopic CU coordination benefit. As described by (22), sufficient patience, i.e. $\beta > \underline{\beta}_{Flex}(\tau)$, implies the farsighted nature of the FTA flexibility benefit dominates the myopic CU coordination benefit. Two reasons explain why $\underline{\beta}_{Flex}(\tau)$ remains constant for $\tau > \bar{\tau}_{IN}^{CU}$. First, $t_{FTA} < t_{CU} < \bar{\tau}_{IN}^{CU}$ implies a tariff binding above $\bar{\tau}_{IN}^{CU}$ does not bind the applied tariffs of FTA nor CU insiders. Second, while $\bar{\tau}_{IN}^{CU} < t_{\varnothing}$ implies a tariff binding above $\bar{\tau}_{IN}^{CU}$ could bind the FTA and CU outsider, they have the same optimal applied tariff. Thus, the myopic CU coordination benefit and the FTA flexibility benefit underlying $\underline{\beta}_{Flex}(\tau)$ in (22) remain constant for tariff bindings above $\bar{\tau}_{IN}^{CU}$.

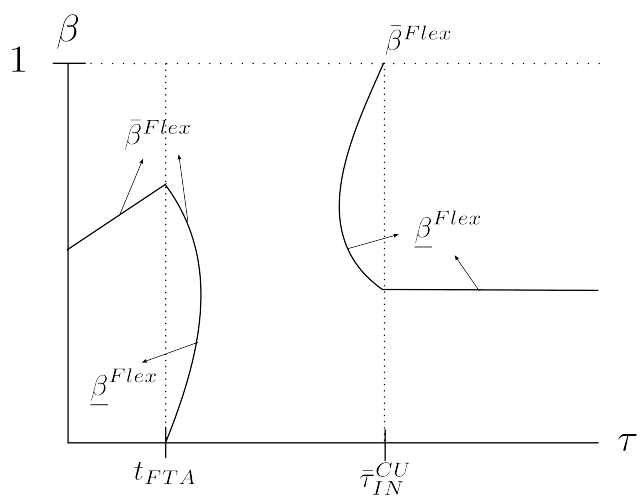

Figure 7: Flexibility benefit of FTAs versus coordination benefits of CUs

While only the myopic CU coordination benefit exists when $\tau > \bar{\tau}_{IN}^{CU}$, only the farsighted CU coordination benefit exists once $\tau < t_{FTA}$. Here, the tariff binding binds FTA and CU insiders and outsiders and, hence, eliminates the myopic CU coordination benefit: $\Delta w_i \left(g_{ij}^{CU} - g_{ij}^{FTA}\right) = 0$. Moreover, $\beta \in \left(\underline{\beta}^{Flex}(\tau), \bar{\beta}^{Flex}(\tau)\right)$ reduces to $\beta < \bar{\beta}^{Flex}(\tau) = \bar{\beta}_{IN}(\tau)$ (see (19)). Nevertheless, the relatively strict tariff binding generates the CU exclusion incentive $\Delta w_i \left(g_{ij}^{CU} - g^{FT}\right) > 0$. In turn, the farsighted CU coordination benefit emerges whereby, unlike FTA insiders who can precipitate global free trade by exploiting the FTA flexibility benefit and becoming the hub, CU formation carries veto power over subsequent CU expansion. Now $\bar{\beta}^{Flex}(\tau)$ balances the FTA flexibility benefit and farsighted CU

coordination benefit with a sufficiently patient country preferring CU formation. Moreover, further tariff binding liberalization strengthens the farsighted CU coordination benefit by increasing the concessions extracted from the CU outsider and weakens the FTA flexibility benefit by shrinking the preference margin. In turn, tariff binding liberalization makes CU formation more attractive by shrinking $\bar{\beta}^{Flex}(\tau)$.

In the bullet shaped regions, where bindings lie just above t_{FTA} or just below $\bar{\tau}_{IN}^{CU}$, both the myopic and farsighted CU coordination benefits exist. Thus, sufficiently patient and sufficiently myopic countries prefer CU formation over being the FTA insider and then the hub on the path to global free trade. As τ falls just below $\bar{\tau}_{IN}^{CU}$, the myopic CU coordination and FTA flexibility benefits remain constant because the tariff binding does not yet bind the tariffs of PTA insiders nor spokes. However, the CU exclusion incentive strengthens because the tariff binding now binds the CU outsider and hence delivers concessions to CU insiders. This stronger farsighted CU coordination benefit increases the attractiveness of CU formation and, indeed, countries eventually prefer CU formation regardless of the discount factor. Once $\tau \in (t_{FTA}, t_{CU})$, the FTA flexibility remains constant but the myopic CU coordination benefit weakens because the tariff binding now binds CU insiders. As τ nears t_{FTA}, FTA formation again becomes attractive for an intermediate range of the discount factor and becomes more attractive as the myopic CU coordination benefit continues weakening and eventually disappears.

With an understanding of the trade off between the FTA flexibility benefit and CU coordination benefits in place, Proposition 3 characterizes the equilibrium path of networks when countries endogenously choose between FTAs and CUs. Here, we let $\underline{\beta}_{Flex}(\tau) \equiv 0$ when the myopic CU coordination benefit disappears (i.e. $\tau < t_{FTA}$) and, similarly, we let $\bar{\beta}^{Flex}(\tau) \equiv 1$ when the CU exclusion incentive disappears (i.e. $\tau > \bar{\tau}_{IN}^{CU}$).

Proposition 3 *In equilibrium, FTA formation emerges when $\beta > \bar{\beta}_{OUT}(\tau)$ and $\beta \in \left(\underline{\beta}_{Flex}(\tau), \bar{\beta}^{Flex}(\tau)\right)$ but CU formation emerges otherwise. When FTA formation emerges in equilibrium, the equilibrium path of networks is $g_{\varnothing} \to g_{ab}^{FTA} \to g_a^H \to g^{FT}$ if $\tau < \tilde{\tau}$ but $g_{\varnothing} \to g_{ac}^{FTA} \to g_a^H \to g^{FT}$ if $\tau \geq \tilde{\tau}$. When CU formation emerges in equilibrium, the equilibrium path of networks is $g_{\varnothing} \to g_{ab}^{CU} \to g^{FT}$ if $\tau > \bar{\tau}_{IN}^{CU}$ but $g_{\varnothing} \to g_{ab}^{CU}$ if $\tau \leq \bar{\tau}_{IN}^{CU}$.*

Figure 8 illustrates Proposition 3. From our above discussion, one may have expected that FTA formation would emerge in equilibrium if and only if $\beta \in \left(\underline{\beta}_{Flex}(\tau), \bar{\beta}^{Flex}(\tau)\right)$. However, the trade off underlying this logic presumes FTA formation leads to global free trade. In contrast, Section 3 showed this only happens if the FTA outsider does not hold a dynamic free riding incentive and FTA insiders do not hold a dynamic exclusion incentive. On one hand, the myopic CU coordination benefit implies an FTA insider cannot simulta-

neously hold a dynamic exclusion incentive and prefer FTA formation over CU formation (i.e. $\bar{\beta}^{Flex}(\tau) \leq \bar{\beta}_{IN}(\tau)$). On the other hand, the FTA outsider may hold a dynamic free riding incentive when an FTA insider prefers FTA formation over than CU formation; indeed, $\underline{\beta}_{Flex}(\tau) < \bar{\beta}_{OUT}(\tau)$ obtains when $\tau > \bar{\tau}_{IN}^{CU}$. Thus, the equilibrium emergence of FTA formation requires not only that the FTA flexibility dominate the CU coordination benefits but also that the FTA outsider does not hold a dynamic free riding incentive. In this case, FTA expansion reaches global free trade.

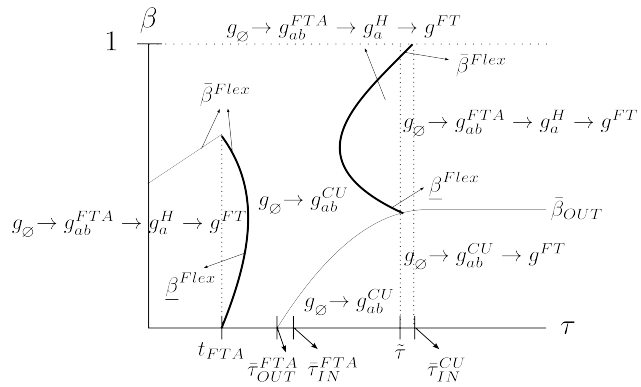

Figure 8: Equilibrium path of networks: PTA game

When FTA formation does not emerge in equilibrium, either because PTA insiders see the CU coordination benefits as dominating the FTA flexibility benefit or because the FTA outsider has a dynamic free riding incentive, PTA insiders form a CU to exploit the CU coordination benefits. If the tariff binding exceeds $\bar{\tau}_{IN}^{CU}$ then CU insiders do not hold a CU exclusion incentive and the CU coordination benefit consists entirely of the myopic CU coordination benefit stemming from the internalization of tariff complementarity. In this case, CU formation expands to global free trade. However, if the tariff binding falls below $\bar{\tau}_{IN}^{CU}$ then CU insiders hold a CU exclusion incentive and exclude the CU outsider from CU expansion to global free trade. Thus, the CU insiders remain permanent CU insiders.

Typically, the PTA literature has viewed the building bloc-stumbling bloc issue as a comparison between PTA formation and multilateral liberalization in terms of whether PTA formation improves or hurts the prospects of global free trade relative to the outcome under multilateral liberalization. However, in practice, multilateral liberalization happens in the short bursts in the background whether it be through various global rounds of negotiations or plurilateral agreements such as the Information Technology Agreement. In contrast, countries are continuously forming PTAs on the surface of the global trade policy landscape and, in doing so, choosing whether to form FTAs or CUs. Thus, an alternative view of the building bloc-stumbling bloc issue, and perhaps a more informative view, would compare

the different types of PTAs in terms of whether a given type of PTA improve the prospects of global free trade relative to the outcome if this type of PTA was banned.

To this end, we introduce the terminology of a "PTA stumbling bloc" and a "PTA building bloc". We say that CUs (FTAs) are a "PTA stumbling bloc" if CU (FTA) formation emerges in equilibrium and does not lead to global free trade yet FTA (CU) formation would lead to global free trade in the absence of CUs (FTAs). Similarly, we say that CUs (FTAs) are a "PTA building bloc" if CU (FTA) formation emerges in equilibrium and leads to global free trade yet FTA (CU) formation would not lead to global free trade in the absence of CUs (FTAs). A comparison of Proposition 3 with Propositions 1-2 reveals the following result.

Corollary 2 *When the tariff binding lies in an intermediate range* $\tau \in \left(t_{FTA}, \bar{\tau}_{IN}^{CU}\right)$*:*

(i) CUs can be a PTA stumbling bloc but not a PTA building bloc

(ii) FTAs can be a PTA building bloc but not a PTA stumbling bloc.

When the tariff binding is sufficiently high, $\tau > \bar{\tau}_{IN}^{CU}$, neither type of PTA is a PTA building bloc or PTA stumbling bloc because both types of PTA lead to global free trade whether they emerge in equilibrium or not. However, CU formation does not expand to global free trade once $\tau < \bar{\tau}_{IN}^{CU}$ (and hence, CUs can never be a PTA building bloc). Nevertheless, FTA formation still expands to global free trade as long as FTA insiders do not hold a dynamic exclusion incentive (i.e. $\beta < \bar{\beta}_{IN}(\tau)$) and the FTA outsider does not hold a dynamic free riding incentive (i.e. $\beta > \bar{\beta}_{OUT}(\tau)$). In this case, CU formation still emerges in equilibrium when the CU coordination benefits dominate the FTA flexibility benefit and, in turn, CUs are PTA stumbling blocs. In contrast, when the FTA flexibility benefit dominates the CU coordination benefits, FTA formation emerges in equilibrium and leads to global free trade even though CU formation would stop short of global free trade. That is, FTAs are PTA building blocs. Finally, noting that the emergence of FTA formation in equilibrium always leads to global free trade, FTAs can never be PTA stumbling blocs. Ultimately, CUs and FTAs play very different roles in terms of the possibility that PTA liberalization reaches global free trade.

6 Conclusion

We investigate the impact of multilateral tariff binding liberalization on the equilibrium extent of Preferential Trade Agreement (PTA) formation. While the 1994 Uruguay Round represents the last successful round of multilateral negotiations, subsequent tariff binding liberalization has taken place on IT products through large plurilateral agreement known as the Information Technology Agreement. Such an agreement has also been discussed as a

template for future plurilateral agreements, including an Environmental Goods Agreement. Thus, multilateral tariff binding liberalization has taken place since the Uruguay Round and is likely to continue taking place moving forward, even in the absence of a successful turn in the currently-stalled Doha Round of negotiations.

Under relatively high tariff bindings, multilateral tariff binding liberalization can facilitate or impede whether FTA expansion reaches global free trade. With relatively high tariff bindings, the intuition revolves around the FTA non-member's dynamic free riding incentive. Here, the FTA non-member weighs a myopic incentive to free ride on the tariff complementarity of FTA members against a farsighted incentive to reach global free trade which eliminates the discrimination practiced by FTA members. When the myopic free riding incentive dominates, continual tariff binding liberalization facilitates FTA expansion to global free trade by constraining the FTA non-members' tariffs on FTA members and, thus, eventually eliminating the myopic free riding incentive. However, when the farsighted incentive for global free trade dominates, continual tariff binding liberalization impedes FTA expansion to global free trade by effectively increasing the FTA non-members' concessions to FTA members and eventually creating an incentive for FTA members to exclude the non-member from subsequent FTA expansion. Ultimately, tariff binding liberalization crucially impacts the incentives governing whether FTA expansion reaches global free trade.

In practice, PTA formation involves countries endogenously choosing whether to form FTAs or CUs. With relatively high tariff bindings, global free trade always emerges in equilibrium whether it be through FTA or CU formation. However, outside these relatively high tariff bindings, CU members want to exclude the CU non-member from CU expansion to global free trade because the tariff bindings effectively force the CU non-member to grant substantial tariff concessions to CU members. Indeed, the equilibrium emergence of CUs can prevent (cannot facilitate) global free trade that would otherwise occur (not occur) through FTAs. In contrast, the equilibrium emergence of FTAs can facilitate (cannot prevent) global free trade that would otherwise not occur (occur) through CUs. FTAs facilitate global free trade when the FTA flexibility benefit of FTAs, i.e. the benefit each FTA member derive from being able to form their own subsequent FTA, dominates the coordination benefits CU members derive from coordination of their external tariffs. Ultimately, tariff binding liberalization casts a negative shadow over the impact of CUs, but a positive shadow over the impact of FTAs, on the prospect for global free trade.

Appendix

A Welfare expressions and optimal tariffs

A.1 Welfare expressions

We begin by reporting welfare levels under different policy regimes as a function of tariffs and comparative advantage. For an arbitrary tariffs vector $\mathbf{t}$=$(t_{ij}, t_{ik}, t_{ji}, t_{jk}, t_{ki}, t_{kj})$, we can write country i's welfare as

$$w_i = \sum_Z CS_i^Z + \sum_Z PS_i^Z + \sum_{z \neq i} t_{iz} x_z^I,$$

where consumer surplus in country i equals

$$\sum_Z CS_i^Z = \frac{1}{2}\left[\left(\alpha - P_i^I\right)^2 + \left(\alpha - P_j^J + t_{ji}\right)^2 + \left(\alpha - P_k^J + t_{ki}\right)^2\right],$$

its producer surplus equals

$$\sum_Z PS_i^Z = \frac{1}{2}\left[\left(P_i^I\right)^2 + (1+\lambda_i)\left(P_j^J - t_{ji}\right)^2 + (1+\lambda_i)\left(P_k^K - t_{ki}\right)^2\right]$$

and the tariff revenue is given by

$$\sum_{z \neq i} t_{iz} x_z^I = t_{ij}[(2+\lambda_j)\, P_j^I - \alpha] + t_{ik}[2+\lambda_k) P_k^I - \alpha]$$

Using the above formulae and the price/optimal tariff levels reported in the text, we can easily calculate welfare levels under all possible trade regimes. To save space, we do not include the algebraic details underlying these straightforward calculations.

B Proofs

Before we present the proofs from the main text, we present an additional proof that will be used in Propositions 2-3.

Lemma 6 *Consider a subgame at g_{ij}^{CU}. The equilibrium outcome in the subgame is*
(i) $g_{ij}^{CU} \rightarrow g^{FT}$ if $\tau > \bar{\tau}_{IN}^{CU}$
(ii) $g_{ij}^{CU} \rightarrow g_{ij}^{CU}$ if $\tau \leq \bar{\tau}_{IN}^{CU}$.

Proof. Lemma 5(ii) implies $V_k\left(g^{FT}\right) > V_k\left(g_{ij}^{CU}\right)$. Moreover, by definition, $V_h\left(g^{FT}\right) > V_h\left(g_{ij}^{CU}\right)$ for $h = i, j$ if and only if $\tau > \bar{\tau}_{IN}^{CU}$. Thus, the first CU insider in the protocol, say i, proposes g^{FT} to j and k, who both accept, if $\tau > \bar{\tau}_{IN}^{CU}$. But, no CU insider accepts a proposal, and hence no CU insider makes a proposal, if $\tau \leq \bar{\tau}_{IN}^{CU}$. Thus, $g_{ij}^{CU} \to g^{FT}$ if $\tau > \bar{\tau}_{IN}^{CU}$ but $g_{ij}^{CU} \to g_{ij}^{CU}$ if $\tau \leq \bar{\tau}_{IN}^{CU}$. ■

We now present proofs of lemmas and propositions from the main text.

Proof of Lemma 1

Using the welfare and optimum tariff levels (when feasible under the given tariff binding) reported in the text, we can show the following for all τ:

$$\begin{aligned}
\Delta w_i(g_{ij}^{FTA} - g_{\varnothing}) &= \Delta w_j(g_{ij}^{FTA} - g_{\varnothing}) > 0 \\
\Delta w_i(g_i^H - g_{ij}^{FTA}) &= \Delta w_i(g_i^H - g_{ik}^{FTA}) > 0 \\
\Delta w_i(g_i^{FT} - g_j^H) &> 0 \text{ and } \Delta w_i(g^{FT} - g_{jk}^{FTA}) > 0
\end{aligned}$$

We also have:

$$\Delta w_i(g_j^H - g_{jk}) > 0 \text{ if and only if } \tau < \bar{\tau}_{OUT}^{FTA} \text{ where } \bar{\tau}_{OUT}^{FTA} \in (t_{FTA}, t_{\varnothing})$$

$$\Delta w_i(g_{ij}^{FTA} - g_{jk}^{FTA}) > 0 \text{ for } \tau < \tilde{\tau} \text{ where } \tilde{\tau} \in \left(\bar{\tau}_{OUT}^{FTA}, t_{\varnothing}\right)$$

and

$$\Delta w_i(g_i^{FT} - g_{ij}^{FTA}) > 0 \text{ if and only if } \tau > \bar{\tau}_{IN}^{FTA} \text{ where } \bar{\tau}_{IN}^{FTA} \in (t_{FTA}, t_{\varnothing}).$$

■

Proof of Lemma 2

Part (i): Using the welfare and optimum tariff levels (when feasible under the given tariff binding) reported in the text, we can show the following:

If $g_{ij}^{FTA} \longrightarrow g_{ij}^{FTA}$, it is immediate from part (i) of Lemma 1 that $\Delta w_i(g_{ij}^{FTA} - g_{\varnothing}) = \Delta w_j(g_{ij}^{FTA} - g_{\varnothing}) > 0$.

If $g_{ij}^{FTA} \to g_i^H \to g^{FT}$, $w_i\left(g_{ij}^{FTA}\right) + \beta w_i\left(g_i^H\right) + \frac{\beta^2}{1-\beta} w_i\left(g^{FT}\right) > w_j\left(g_{ij}^{FTA}\right) + \beta w_j\left(g_i^H\right) + \frac{\beta^2}{1-\beta} w_j\left(g^{FT}\right)$ and $w_j\left(g_{ij}^{FTA}\right) + \beta w_j\left(g_i^H\right) + \frac{\beta^2}{1-\beta} w_j\left(g^{FT}\right) > V_i\left(g_{\varnothing}\right) = V_j\left(g_{\varnothing}\right)$ hold for all β.

Part (ii): For sufficiently high β and relevant τ values over which $g_{ij}^{FTA} \to g_i^H \to g^{FT}$ and $g_{jk}^{FTA} \to g_j^H \to g^{FT}$, using the welfare and optimum tariff levels (when feasible under the given tariff binding) reported in the text, we can show that $w_i\left(g_{ij}^{FTA}\right) + \beta w_i\left(g_i^H\right) + \frac{\beta^2}{1-\beta} w_i\left(g^{FT}\right) > w_i\left(g_{jk}^{FTA}\right) + \beta w_i\left(g_j^H\right) + \frac{\beta^2}{1-\beta} w_i\left(g^{FT}\right)$ always holds.■

Proof of Lemma 3

Lemma 1(i) implies the first spoke in the protocol, say j, proposes an FTA with k who accepts. Thus, $g_i^H \to g^{FT}$.■

Proof of Lemma 4

Given i proposes before j in the protocol, then either i or k proposes in stage 1a. Moreover, Lemma 3 implies $g_z^H \to g^{FT}$ in any subgame at g_z^H. Thus, by definition, $V_i\left(g_i^H\right) > \frac{1}{1-\delta} w_i\left(g_{ij}^{FTA}\right) \Leftrightarrow \beta < \bar{\beta}_{IN}(\tau)$ and $V_k\left(g_i^H\right) > \frac{1}{1-\delta} w_k\left(g_{ij}^{FTA}\right) \Leftrightarrow \beta > \bar{\beta}_{OUT}(\tau)$. Moreover, parts (i) and (iv) of 1 imply $w_i\left(g_i^H\right) > w_i\left(g^{FT}\right) > w_i\left(g_j^H\right)$ so that $V_i\left(g_i^H\right) > V_i\left(g_j^H\right)$.

First, suppose $\beta \in \left(\bar{\beta}_{OUT}(\tau), \bar{\beta}_{IN}(\tau)\right)$. Then, in stage 1a, i (or k) proposes an FTA with k (or i) and k (or i) accepts. Hence, the equilibrium outcome in the subgame is $g_{ij}^{FTA} \to g_i^H$. Second, suppose $\beta < \bar{\beta}_{OUT}(\tau)$. Then, k rejects any FTA proposal received from i or j and chooses to make no proposal as the proposer. Hence, the equilibrium outcome in the subgame is $g_{ij}^{FTA} \to g_{ij}^{FTA}$. Third, suppose $\beta > \bar{\beta}_{IN}(\tau)$. Then, i and j choose to make no FTA proposal as the proposer and reject any proposal received from k. Hence, the equilibrium outcome in the subgame is $g_{ij}^{FTA} \to g_{ij}^{FTA}$.■

Proof of Proposition 1

Note throughout that country b (c) receives an arbitrarily small non-economic benefit $\varepsilon > 0$ from FTA formation with country a rather than country c (b) and country a receives an arbitrarily small non-economic benefit $\varepsilon > 0$ from FTA formation with country b rather than country c. Moreover, Lemmas 3 and 4 describe the equilibrium transitions from subgames at the FTA insider-outsider networks and hub-spoke networks. In particular, letting i be the most attractive FTA insider, $g_{ij}^{FTA} \to g_i^H \to g^{FT}$ if and only if $\beta \in \left(\bar{\beta}_{OUT}(\tau), \bar{\beta}_{IN}(\tau)\right)$ but $g_{ij}^{FTA} \to g_{ij}^{FTA}$ otherwise.

Stage 2. First, suppose $\beta \notin \left(\bar{\beta}_{OUT}(\tau), \bar{\beta}_{IN}(\tau)\right)$. Then, Lemma 4 implies $g_{ij}^{FTA} \to g_{ij}^{FTA}$ and $V_z\left(g_{ij}^{FTA}\right) > V_z\left(g_{\varnothing}\right)$ for $z = i, j$ by Lemma 1(i). Second, suppose $\beta \in \left(\bar{\beta}_{OUT}(\tau), \bar{\beta}_{IN}(\tau)\right)$. Then, Lemmas 3 and 4 imply $g_{ij}^{FTA} \to g_i^H \to g^{FT}$ where country i is the more attractive FTA insider. In turn, $V_z\left(g_{ij}^{FTA}\right) > V_z\left(g_{\varnothing}\right)$ for $z = i, j$ either by Lemma 1(i) or Lemma 2(i). Thus, country b proposes FTA formation with country a, who accepts, when $\beta \notin \left(\bar{\beta}_{OUT}(\tau), \bar{\beta}_{IN}(\tau)\right)$ given the non-economic benefits but proposes FTA formation with country c, who accepts, when $\beta \in \left(\bar{\beta}_{OUT}(\tau), \bar{\beta}_{IN}(\tau)\right)$ given parts (i) and (iv) of 1 imply $w_b\left(g_b^H\right) > w_b\left(g^{FT}\right) > w_b\left(g_a^H\right)$ so that $V_b\left(g_{bc}^{FTA}\right) > V_b\left(g_{ab}^{FTA}\right)$.

Stage 1b. Note the equilibrium outcome in Stage 2 and that $V_a\left(g_{ab}^{FTA}\right) = V_a\left(g_{ac}^{FTA}\right) > V_a\left(g_{\varnothing}\right)$. First, suppose $\beta \notin \left(\bar{\beta}_{OUT}(\tau), \bar{\beta}_{IN}(\tau)\right)$. Then, country a proposes FTA formation with country b, who accepts, if country c rejected country a's proposal in Stage 1a. But, given the non-economic benefits, country a makes no proposal if country b rejected country a's proposal in Stage 1a.

Second, suppose $\beta \in \left(\bar{\beta}_{OUT}(\tau), \bar{\beta}_{IN}(\tau)\right)$. Then, given Lemma 2(ii), country a proposes FTA formation with country c, who accepts given the non-economic benefits, if country b rejected country a's proposal in Stage 1a. But, given the outcome of g_{bc}^{FTA} in Stage 2 implies country b will reject an FTA proposal from country a, country a makes no proposal to country b if country c rejected country a's proposal in Stage 1a.

Stage 1a. First, suppose $\beta \in \left(\bar{\beta}_{OUT}(\tau), \bar{\beta}_{IN}(\tau)\right)$ noting that $V_i\left(g_{ij}^{FTA}\right) > V_i\left(g_{jk}^{FTA}\right)$ by 2(ii). Then, the non-economic benefits imply country a proposes an FTA with country b who accepts given the equilibrium outcome of g_{ac}^{FTA} in Stage 1b upon its rejection in Stage 1a. Second, suppose $\beta \notin \left(\bar{\beta}_{OUT}(\tau), \bar{\beta}_{IN}(\tau)\right)$. Then, the non-economic benefits imply country a proposes an FTA with country b who, given the equilibrium outcome of g_{ab}^{FTA} in Stage 1b, accepts. Thus, the equilibrium path of networks is $g_{\varnothing} \to g_{ab}^{FTA} \to g_a^H \to g^{FT}$ if $\beta \in \left(\bar{\beta}_{OUT}(\tau), \bar{\beta}_{IN}(\tau)\right)$ but $g_{\varnothing} \to g_{ab}^{FTA}$ if $\beta \notin \left(\bar{\beta}_{OUT}(\tau), \bar{\beta}_{IN}(\tau)\right)$.■

Proof of Lemma 5

Using the welfare and optimum tariff levels (when feasible under the given tariff binding) reported in the text, we can show the following for all τ:

$$\Delta w_i(g_{ij}^{CU} - g_{\varnothing}) > 0 \ \Delta w_k(g_{ij}^{CU} - g_{\varnothing}) < 0 \text{ and } \Delta w_k(g^{FT} - g_{ij}^{CU}) > 0.$$

We also have:

$$\Delta w_i(g^{FT} - g_{ij}^{CU}) > 0 \text{ if and only if } \tau > \bar{\tau}_{IN}^{CU} \text{ where } \bar{\tau}_{IN}^{CU} \in (t_{CU}, t_{\varnothing})$$

and

$$\Delta w_i(g_{ij}^{CU} - g_{ij}^{FTA}) > 0 \text{ for } \tau > t_{FTA}.$$

■

Proof of Proposition 2

Note throughout that country b (c) receives an arbitrarily small non-economic benefit $\varepsilon > 0$ from CU formation with country a rather than country c (b) and country a receives an arbitrarily small non-economic benefit $\varepsilon > 0$ from CU formation with country b rather than country c. Moreover, Lemma 6 describes the equilibrium transitions from CU insider-outsider networks with $g_{ij}^{CU} \to g^{FT}$ if $\tau > \bar{\tau}_{IN}^{CU}$ but $g_{ij}^{CU} \to g_{ij}^{CU}$ if $\tau \leq \bar{\tau}_{IN}^{CU}$.

Stage 2. Lemma 5(i) and, given the veto power of CU members, Lemma 6 imply that $V_i\left(g_{ij}^{CU}\right) > V_i(g_{\varnothing})$ regardless of the equilibrium transition in the subgames at g_{ij}^{CU}. Thus, given the non-economic benefits, country b proposes a CU with country a and country a accepts.

Stage 1b. Given the equilibrium outcome in Stage 2 and the non-economic benefits, country a makes no proposal to c if country b rejected country a's proposal in Stage 1a but country a proposes a CU with country b, and country b accepts, if country c rejected country a's proposal in Stage 1a.

Stage 1a. Given g_{ab}^{CU} emerges in Sage 1b or Stage 2, country a proposes a CU with country b who accepts. Thus, the equilibrium path of networks is $g_{\varnothing} \rightarrow g_{ab}^{CU} \rightarrow g^{FT}$ if $\tau > \bar{\tau}_{IN}^{CU}$ but $g_{\varnothing} \rightarrow g_{ab}^{CU}$ if $\tau \leq \bar{\tau}_{IN}^{CU}$.■

Proof of Proposition 3

Note throughout that country b (c) receives an arbitrarily small non-economic benefit $\varepsilon > 0$ from FTA formation with country a rather than country c (b) and country a receives an arbitrarily small non-economic benefit $\varepsilon > 0$ from FTA formation with country b rather than country c. Moreover, Lemmas 3, 4 and 6 describe the equilibrium transitions from, respectively, hub-spoke, FTA insider-outsider and CU insider-outsider networks.

First, suppose $\beta \notin \left(\bar{\beta}_{OUT}(\tau), \bar{\beta}_{IN}(\tau)\right)$ so that $g_{ij}^{FTA} \rightarrow g_{ij}^{FTA}$.

Stage 2. The veto power held by CU insiders over CU expansion together with parts (i) and (iii) of Lemma 5 and Lemma 1(i) implies $V_i\left(g_{ij}^{CU}\right) \geq V_i\left(g_{ij}^{FTA}\right) > V_i\left(g_{\varnothing}\right)$. Thus, given the non-economic benefits, country b proposes CU formation with a who accepts.

Stage 1b. As in Stage 2, $V_i\left(g_{ij}^{CU}\right) \geq V_i\left(g_{ij}^{FTA}\right) > V_i\left(g_{\varnothing}\right)$. Moreover, given the equilibrium outcome in Stage 2 of g_{ab}^{CU}, country b accepts a CU proposal from country a. Thus, if country c rejected country a's proposal in Stage 1a so that country a can propose to country b, the non-economic benefits imply country a proposes CU formation with country b who accepts. In contrast, given the equilibrium outcome in Stage 2 of g_{ab}^{CU}, country a makes no proposal if country b rejected country a's proposal in Stage 1a.

Stage 1a. Given the equilibrium outcome in Stage 1b or Stage 2 and the logic in these stages, country a proposes CU formation with country b who accepts. Thus, the equilibrium path of networks when $\beta \notin \left(\bar{\beta}_{OUT}(\tau), \bar{\beta}_{IN}(\tau)\right)$ is $g_{\varnothing} \rightarrow g_{ab}^{CU}$ if $\tau \leq \bar{\tau}_{IN}^{CU}$ but $g_{\varnothing} \rightarrow g_{ab}^{CU} \rightarrow g^{FT}$ if $\tau > \bar{\tau}_{IN}^{CU}$.

Second, suppose $\beta \in \left(\bar{\beta}_{OUT}(\tau), \bar{\beta}_{IN}(\tau)\right)$ so that $g_{ij}^{FTA} \rightarrow g_{i}^{H} \rightarrow g^{FT}$ where country i is the more attractive FTA insider in terms of non-economic benefits.

Stage 2. Lemma 1(i) combined with Lemma 2, Lemma 5(iii) and the veto power of CU insiders over CU expansion implies that $V_b\left(g_{ab}^{CU}\right) \geq V_b\left(g_{ab}^{FTA}\right) > V_b\left(g_{\varnothing}\right)$. Moreover, $V_b\left(g_{bc}^{FTA}\right) > V_b\left(g_{bc}^{CU}\right)$ if and only if $\beta \in \left(\underline{\beta}^{Flex}(\tau), \bar{\beta}^{Flex}(\tau)\right)$ but $V_c\left(g_{bc}^{FTA}\right) > V_c\left(g_{\varnothing}\right)$ by Lemma 2(i). Thus, given $V_b\left(g_{bc}^{CU}\right) = V_b\left(g_{ab}^{CU}\right)$, country b proposes FTA formation with country c, who accepts, if $\beta \in \left(\underline{\beta}^{Flex}(\tau), \bar{\beta}^{Flex}(\tau)\right)$. But, given the non-economic benefits, country b proposes CU formation with country a, who accepts, if $\beta \notin \left(\underline{\beta}^{Flex}(\tau), \bar{\beta}^{Flex}(\tau)\right)$.

Stage 1b. Suppose country c rejected country a's proposal in Stage 1a so that country a can propose to country b. Given the equilibrium outcome in Stage 2, country b will only accept country a's proposal if country a proposes CU formation and will only accept a CU proposal if $\beta \notin \left(\underline{\beta}^{Flex}(\tau), \bar{\beta}^{Flex}(\tau)\right)$. Indeed, following similar logic to Stage 2, country a proposes CU formation with country b, who accepts, when $\beta \notin \left(\underline{\beta}^{Flex}(\tau), \bar{\beta}^{Flex}(\tau)\right)$. But, given the equilibrium outcome in Stage 2 of g_{bc}^{FTA} when $\beta \in \left(\underline{\beta}^{Flex}(\tau), \bar{\beta}^{Flex}(\tau)\right)$, country a makes no proposal when $\beta \in \left(\underline{\beta}^{Flex}(\tau), \bar{\beta}^{Flex}(\tau)\right)$.

Now suppose country b rejected country a's proposal in Stage 1a so that country a can propose to country c. Let $\beta \in \left(\underline{\beta}^{Flex}(\tau), \bar{\beta}^{Flex}(\tau)\right)$. Then, $V_i\left(g_{ij}^{FTA}\right) > V_i\left(g_{ij}^{CU}\right)$ where country i is more attractive than country j based on non-economic benefits. Thus, given the equilibrium outcome in Stage 2 of g_{bc}^{FTA} and the non-economic benefits, Lemma 2(ii) implies country a proposes FTA formation with country c who accepts. Now let $\beta \notin \left(\underline{\beta}^{Flex}(\tau), \bar{\beta}^{Flex}(\tau)\right)$. Then, $V_i\left(g_{ij}^{CU}\right) > V_i\left(g_{ij}^{FTA}\right)$ where country i is more attractive than country j based on non-economic benefits. Thus, given the equilibrium outcome in Stage 2 of g_{ab}^{CU} and the non-economic benefits, country a proposes CU formation with country b who accepts.

Stage 1a. Given the equilibrium outcome in Stage 1b and the non-economic benefits, country a proposes a CU with country b, who accepts, when $\beta \notin \left(\underline{\beta}^{Flex}(\tau), \bar{\beta}^{Flex}(\tau)\right)$ but, given Lemma 2(ii), country a proposes an FTA with country b, who accepts, when $\beta \in \left(\underline{\beta}^{Flex}(\tau), \bar{\beta}^{Flex}(\tau)\right)$.

Thus, the equilibrium path of networks when $\beta \in \left(\bar{\beta}_{OUT}(\tau), \bar{\beta}_{IN}(\tau)\right)$ is (i) $g_{\varnothing} \to g_{ab}^{FTA} \to g_i^H \to g^{FT}$ if $\beta \in \left(\underline{\beta}^{Flex}(\tau), \bar{\beta}^{Flex}(\tau)\right)$, (ii) $g_{\varnothing} \to g_{ab}^{CU}$ if $\beta \notin \left(\underline{\beta}^{Flex}(\tau), \bar{\beta}^{Flex}(\tau)\right)$ and $\tau \leq \bar{\tau}_{IN}^{CU}$ and (iii) $g_{\varnothing} \to g_{ab}^{CU} \to g^{FT}$ if $\beta \notin \left(\underline{\beta}^{Flex}(\tau), \bar{\beta}^{Flex}(\tau)\right)$ and $\tau > \bar{\tau}_{IN}^{CU}$. Finally, note that $\bar{\beta}_{IN}(\tau) \geq \bar{\beta}^{Flex}(\tau)$ and thus the constraint of Lemma 4 that $g_{ij}^{FTA} \to g_i^H \to g^{FT}$ requires $\beta < \bar{\beta}_{IN}(\tau)$ does not bind on the equilibrium path.■

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